

AI for IoMT security: a comprehensive survey of intrusion detection and system architectures

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ABSTRACT

Recent advances in the Internet of Medical Things (IoMT) have significantly improved data processing and patient care within Smart Healthcare systems. However, these developments have also expanded the surface of potential cyber threats targeting sensitive medical infrastructures. To address these challenges, a variety of security approaches both traditional and Artificial Intelligence (AI)-based have been proposed to strengthen the resilience of IoMT environments. In particular, Machine Learning (ML) and Deep Learning (DL) techniques have demonstrated strong capabilities in detecting and mitigating abnormal behaviors and malicious activities. This paper provides a comprehensive survey of recent AI-driven methods applied to IoMT security, with a particular focus on intrusion detection systems (IDS), the availability and characteristics of public datasets, and architectural considerations for deploying security solutions across Cloud, Fog, and Edge computing layers. The paper also discusses legal and ethical concerns related to data protection in healthcare contexts. Finally, the study outlines open challenges and future research directions for developing robust, adaptive, and trustworthy security frameworks in the IoMT ecosystem.

1. Introduction

Healthcare systems rooted in traditional care models are increasingly overwhelmed, lacking the agility to manage diverse medical conditions, complex treatment protocols, and patients' immediate needs. This pressure is amplified by growing patient volumes and constrained human and material resources [1]. In this context, the adoption of Internet of Things (IoT) technologies in the healthcare sector has emerged as a promising approach. By relying on medical devices connected to online services, healthcare professionals and patients can exchange data seamlessly, enabling enhanced responsiveness, continuous monitoring, and improved quality of care in real time [2]. When IoT technologies are applied to the medical domain, they are commonly referred to as the IoMT, also known as the Internet of Health Things (IoHT) or Health 4.0 [3]. As illustrated in Fig. 1, the IoMT ecosystem encompasses a wide range of deployment contexts, including in-body, in-home, in-hospital, and in-ambulance environments. Each context represents a critical point of care where connected medical devices contribute to real-time data collection, monitoring, and decision-making. In-body applications refer to implantable or ingestible devices, such as smart pacemakers, biosensors, and connected drug delivery systems, which continuously monitor internal physiological parameters and transmit data remotely. In-home applications involve wearable and stationary devices (e.g., smart blood pressure monitors, glucose sensors, and telehealth platforms) that support remote patient

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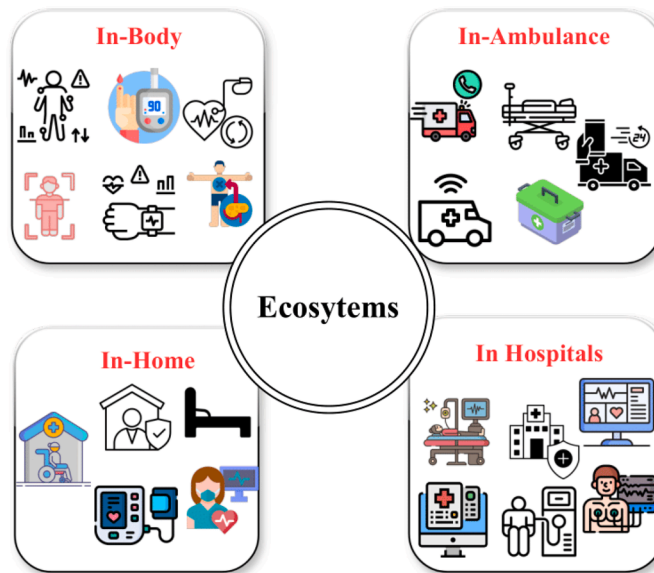


Fig. 1. Healthcare IoT deployment ecosystems.

monitoring and chronic disease management [4]. In-hospital technologies include interconnected systems such as smart infusion pumps, automated medication dispensers, vital sign monitors, and centralized patient dashboards, which optimize resource allocation and facilitate real-time communication between caregivers [5]. In-ambulance applications enable the transmission of critical health data from emergency transport vehicles to receiving medical facilities, using mobile ECG monitors, portable ventilators, and GPS-integrated systems to reduce response times and support pre-arrival preparation [6].

The IoMT architecture is inherently heterogeneous. It involves multiple layers and components, including resource-constrained hardware, diverse communication technologies and protocols, and IoT applications for data visualization and decision support [7]. IoMT devices continuously collect and transmit data, enabling remote monitoring, early detection of health issues, and timely intervention, especially for chronic conditions [8]. These capabilities contribute to personalized treatment plans, optimized disease management, reduced hospital readmissions, and responsive alert systems. Recent market analyses confirm the rapid expansion of IoMT solutions, with the global market expected to grow from USD 60.03 billion in 2024 to USD 814.28 billion by 2032 [9]. This growth reflects the increasing deployment of IoMT applications for real-time monitoring, historical data analysis, and automated report generation through IoT platforms. However, despite these benefits, the IoMT infrastructure presents significant cybersecurity challenges. Many connected medical devices are designed with limited computational resources and often without robust, built-in security mechanisms [10]. The overall ecosystem, composed of heterogeneous devices, wireless networks, cloud platforms, and third-party services, becomes a prime target for cyberattacks. The risks include unauthorized access to sensitive patient data, manipulation or disruption of medical devices, tampering with sensor readings, and service outages that may directly endanger patient safety [11]. The presence of measurement noise, device malfunction, and even deliberate adversarial perturbations further complicates the reliable detection of anomalies and intrusions in IoMT environments. In this context, information and communication security is a top priority to ensure the confidentiality, integrity, and availability of IoMT systems. Traditional security solutions such as antivirus software, firewalls, and signature-based Intrusion Detection Systems (IDS) are no longer sufficient to cope with the sophistication, variability, and real-time constraints of modern IoMT cyber threats [12]. There is a growing need for intelligent, adaptive, and context-aware defense mechanisms capable of learning complex patterns from high-dimensional, noisy, and often imbalanced medical data. Artificial Intelligence (AI) and Machine Learning (ML) techniques, including Deep Learning (DL) and, more recently, Federated Learning (FL) and Explainable AI (XAI), have therefore become key enablers for next-generation IDS and security architectures tailored to IoMT ecosystems.

Several survey papers have reviewed IoT or healthcare security and others have discussed AI-based IDS in general network environments. However, only a limited number of studies specifically focus on AI-driven intrusion detection for IoMT while jointly considering (i) the unique characteristics and constraints of medical devices and data, (ii) the architectural design of IoMT systems, and (iii) practical deployment challenges such as noise, heterogeneity, privacy, and resource limitations. Existing surveys often either concentrate on high-level security requirements without a detailed analysis of AI-based approaches, or address generic IoT scenarios without discussing IoMT-specific constraints and use cases.

This paper is motivated by the increasing number and sophistication of cyberattacks targeting connected healthcare infrastructures, combined with the growing adoption of AI-based security mechanisms in medical environments. At the same time, healthcare providers must comply with strict privacy, safety, and regulatory requirements, which makes the design and deployment of intrusion detection systems in IoMT particularly challenging. Despite the critical nature of these systems, there is still a lack of a focused and

structured survey that connects IoMT architectures, AI-based IDS techniques, FL for privacy-preserving collaboration, and explainable AI for transparency and trust.

To address this gap, this survey provides a comprehensive and structured review of AI-based security mechanisms for IoMT, with a particular emphasis on intrusion detection and system architectures. The main contributions of this paper can be summarized as follows:

- 1) It presents a consolidated overview of IoMT architectures and application ecosystems, highlighting the specific security requirements and threat landscape associated with medical devices and services.
- 2) It proposes a taxonomy of AI-based intrusion detection approaches for IoMT, covering traditional ML, DL, FL, and XAI techniques, and systematically analyzes how these methods are integrated into different IoMT architectural layers.
- 3) It conducts a comparative analysis of existing AI-driven IoMT IDS solutions, summarized in [Table 1](#), focusing on datasets, learning models, evaluation metrics, identified limitations open challenges, and outlines promising research directions for future work.

The scope of this survey is explicitly centered on AI-based intrusion detection and related security architectures in IoMT environments. Cryptographic mechanisms, access control models, and blockchain-based solutions are mentioned only when they directly interact with IDS or influence system design. By clarifying this scope, the paper is intended to serve both researchers and practitioners who seek to understand, design, or deploy intelligent IDS in healthcare IoT infrastructures.

2. Related work and contributions

This section analyzes existing review and survey articles on intrusion detection in the IoMT in order to highlight the added value of the present work. It also provides an objective assessment of the strengths and weaknesses of previous studies. [Table 1](#) summarizes, for each related survey, its main focus, the main limitations identified in the context of IoMT security, and how these limitations are addressed in this survey.

Si-Ahmed et al. [13] classified attacks based on different levels of the IoMT and focused particularly on data collection attacks at the perception layer, which is responsible for gathering patient data. The study also discussed the transmission and storage layers, associated with data visualization applications and databases, and provided examples of ML modules for attack detection. The analysis is grounded on a generic three-layer IoMT architecture consisting of the data acquisition layer, the personal server layer, and the medical server layer. In contrast, the present survey adopts a four-layer architecture, comprising the perception, network, cloud, and application layers. Each layer is examined individually, with a specific focus on its associated threats and vulnerabilities, relevant attack examples, and potential countermeasures. This additional granularity aims to better capture the complexity of modern IoMT environments and to support the design of more targeted and effective security solutions.

Carlos Pinto Neto et al. [14] provided an overview of IoT applications in healthcare systems and explored the use of ML techniques to enhance IoT security in this context. The study also presented several publicly available datasets designed for IoT security in medical environments and identified a range of attacks threatening the confidentiality, integrity, and availability (CIA) of healthcare systems, together with tools for data generation and collection. However, the dataset review remains limited and is largely oriented toward generic IoT. The present survey extends this perspective by focusing specifically on IoMT, introducing additional and more recent datasets that were not included in Neto et al. [14], and offering a broader and more up-to-date view of available resources for AI-based IoMT IDS research.

Naghieb et al. [15] presented a systematic review of intrusion detection mechanisms in IoMT systems. The study categorized key AI models, datasets, and detection approaches, and highlighted the main challenges that must be addressed to improve IDS performance in this domain. It also provided an overview of MIoT architecture by classifying devices according to their corresponding IoMT layers, examined the security and privacy requirements specific to IoMT, and classified various types of attacks based on the targeted layer. The present survey complements this work by covering a broader range of recent studies related to AI-based intrusion detection, including both ML- and DL-based approaches, as well as FL- and XAI-enabled IDS. In addition, the analysis provides a more detailed discussion of attack types affecting IoMT systems and their impact on architectural components, as well as of datasets, attack simulation tools, and deployment aspects.

Zachos et al. [16] proposed an anomaly-based intrusion detection system to secure IoMT networks, combining a hybrid approach deployed on Raspberry Pi devices. The architecture integrates a Local Detection Engine, which monitors device behavior (CPU, memory, and disk usage), and a Central Detection Engine responsible for analyzing network traffic. This lightweight solution offers promising perspectives for the protection of connected medical devices and emphasizes the need for extended testing in more complex network environments and further model optimization to reduce computational overhead. Unlike [16], which focuses on a single implementation and testbed, the present survey compiles and analyzes multiple studies that examine the practical deployment of IDS solutions in healthcare settings, and uses such implementations as illustrative examples within a broader comparative analysis of design choices and open challenges. Overall, this article adopts a different orientation compared with the above works.

First, it provides a detailed explanation of a four-layer IoMT architecture and discusses the specific security challenges and types of attacks associated with each layer.

Second, it presents an in-depth study of the datasets available in the fields of IoMT and IoT for IDS development, and offers additional information on the most prevalent attack types in the medical sector, including an overview of real-world incidents reported in recent years.

Third, it is, to the best of current knowledge, among the first surveys to jointly classify and associate cyberattacks with both the targeted IoMT layer and the compromised CIA security properties, relying on clear and transparent visualization methods.

Table 1

Comparative analysis of existing surveys on IoMT security and their limitations with respect to the present work.

Study	Main focus	Main limitations identified in this paper	How these limitations are addressed in this survey
Si-Ahmed et al. [13]	Classification of attacks and security challenges over a three-layer IoMT / MIoT architecture, with emphasis on data collection attacks at the perception layer.	Relies on a generic three-layer architecture with limited granularity; does not provide a detailed taxonomy of AI-based IDS (ML, DL, FL, XAI); limited coverage of datasets and practical deployment aspects.	Adopts a four-layer IoMT architecture with finer granularity; introduces a taxonomy of AI-based IDS including ML, DL, FL and XAI; extends the analysis to datasets, CIA-based attack mapping, attack simulation tools and deployment aspects.
Neto et al. [14]	Overview of IoT applications in healthcare and available IoT security datasets for medical environments.	Focuses on generic IoT rather than IoMT; limited analysis of intrusion detection mechanisms; no mapping of attacks to IoMT layers or CIA properties; FL and XAI are not considered.	Focuses specifically on IoMT; classifies AI-based IDS and attacks per IoMT layer and compromised CIA property; provides an extended and up-to-date taxonomy of IoMT/IoT datasets; discusses the role of FL and XAI in IoMT IDS.
Naghib et al. [15]	Systematic review of intrusion detection mechanisms in IoMT systems, including AI models, datasets and security/privacy requirements.	Mainly concentrates on ML/DL-based IDS; limited coverage of FL and XAI; does not elaborate on attack simulation tools, next-generation SOC perspectives or real-world deployment scenarios.	Complements the systematic view with an explicit taxonomy including FL- and XAI-enabled IDS; adds a structured discussion on datasets, attack simulation tools and next-generation SOC-oriented IoMT security architectures and deployments.
Zachos et al. [16]	Design and evaluation of a specific anomaly-based IDS architecture for IoMT using Raspberry Pi devices (local and central detection engines).	Not a survey; focuses on a single implementation and testbed; does not provide a broad comparison of AI-based IoMT IDS, nor a dataset taxonomy or attack classification across architectures.	Positions this implementation within a broader landscape of AI-driven IoMT IDS; uses it as one of several deployment-oriented case studies in the comparative analysis and discussion of open challenges.
This survey	Comprehensive survey of AI-based IDS and security architectures for IoMT, including ML, DL, FL and XAI.	–	Provides a unified four-layer IoMT architecture; a taxonomy of AI-based IDS (ML, DL, FL, XAI); an in-depth analysis of IoMT/IoT datasets; a CIA- and layer-based classification of attacks; a discussion of attack simulation tools, deployment aspects and next-generation SOC-oriented IoMT security.

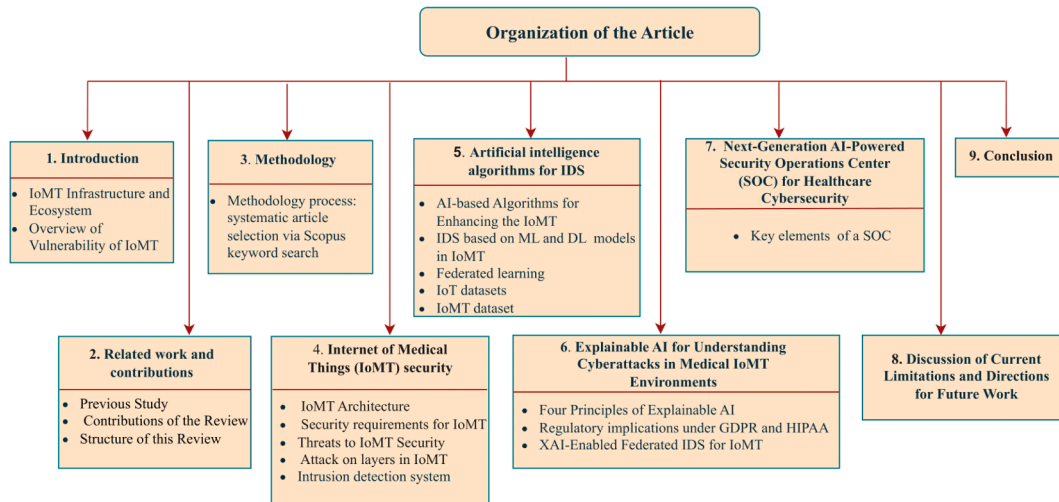


Fig. 2. Illustrates the different sections covered in this literature review.

Furthermore, the survey describes a process linking AI techniques with IDS for anomaly detection, covering key steps such as data preprocessing, feature selection, training, validation and testing of algorithms, and performance evaluation. It also reviews different IDS types and provides examples of open-source IDS tools relevant to IoMT security.

Finally, the article discusses the emerging role of FL, Explainable AI (XAI), and next-generation AI-powered Security Operations Centers (SOC) in real-time threat detection and coordinated defense for healthcare infrastructures.

This work is intended to serve as a reference for researchers and practitioners focusing on security challenges in IoMT. It provides a comprehensive assessment of system vulnerabilities and cyber threats, with particular emphasis on AI-driven methodologies and IDS. The main contributions can be summarized as follows:

- A detailed analysis of the IoMT architecture based on a four-layer model, used as a foundation for mapping threats, attacks, and security requirements and for structuring a new taxonomy of IDS tailored to IoMT.
- A classification of threats and attacks targeting each layer of the IoMT architecture, including their impact on CIA properties and illustrative real-world incidents in healthcare environments.
- An innovative classification of intrusion detection models dedicated to IoMT, considering response strategies, data sources, architectural choices, and AI algorithms, together with an in-depth analysis of IoMT/IoT datasets used for IDS design and evaluation.
- A dedicated discussion of FL and XAI as key enablers for privacy-preserving and trustworthy IoMT intrusion detection, highlighting their roles in collaborative IDS design, model deployment, and the interpretation of AI-based decisions in healthcare environments.

The structure and key themes of this survey are illustrated in Fig. 2, which provides a visual roadmap of the main topics covered and shows how the different sections are organized and interconnected.

3. Methodology

This study employs a four-stage methodology to identify and select pertinent articles, with the search strategy detailed in Fig. 3. In the first stage, the articles published between 2014 and June 2025 were retrieved based on the search phrases and keywords. The initial search on the Scopus database, using the query ((“Internet of Medical Things” OR “IoMT” OR “smart healthcare” OR “IoHT” OR “healthcare 4.0” OR “healthcare”) AND (“cybersecurity” OR “IDS”) AND (“ML” OR “DL” OR “machine learning” OR “deep learning” OR “AI”)), yielded 1317 articles. These articles include a variety of publication types such as scholarly journals, conference proceedings, book chapters, technical reports, and special issues. In Stage 2, the search results were filtered by subject area and limited to Computer Science, with the objective of excluding articles from unrelated fields. In Stage 3, filtering by document type was applied, limiting the selection to ‘Conference Papers’ and ‘Articles’ with the objective of focusing on recent research trends and prioritizing publications based on their latest citation counts. In Stage 4, a total of 163 articles were analyzed. To provide insight into the distribution of selected articles by publisher over time, Fig. 4 illustrates the percentage share of publications from major publishers (Elsevier, IEEE, Springer, MDPI) across different time intervals. This visualization highlights the evolution of publishing trends within the field of IoMT and cybersecurity research. To further illustrate the temporal evolution of research activity, Fig. 5 presents the percentage distribution of the selected articles by year. This visualization highlights trends in publication frequency over time, shedding light on the growing interest in IoMT security research.

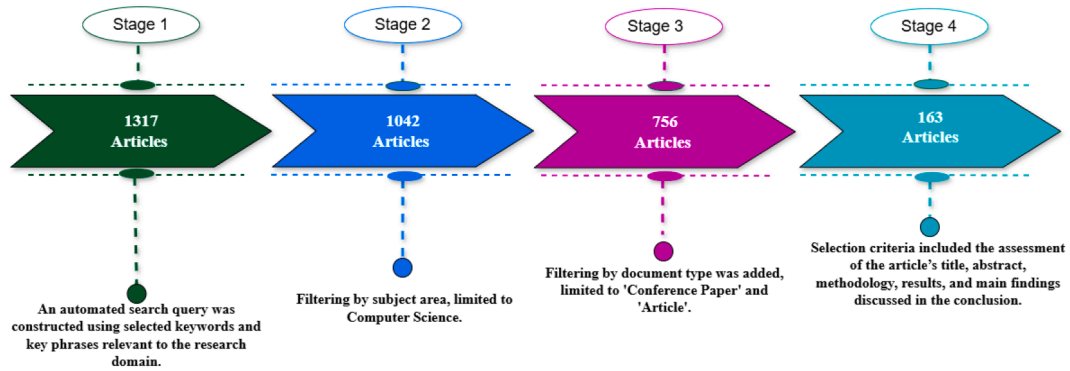


Fig. 3. Article selection and filtering process.

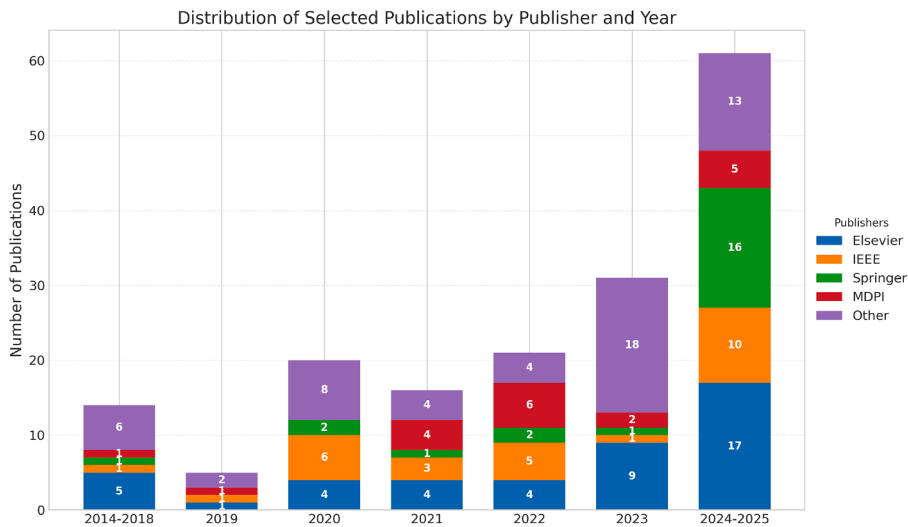


Fig. 4. Annual publication counts by publisher.

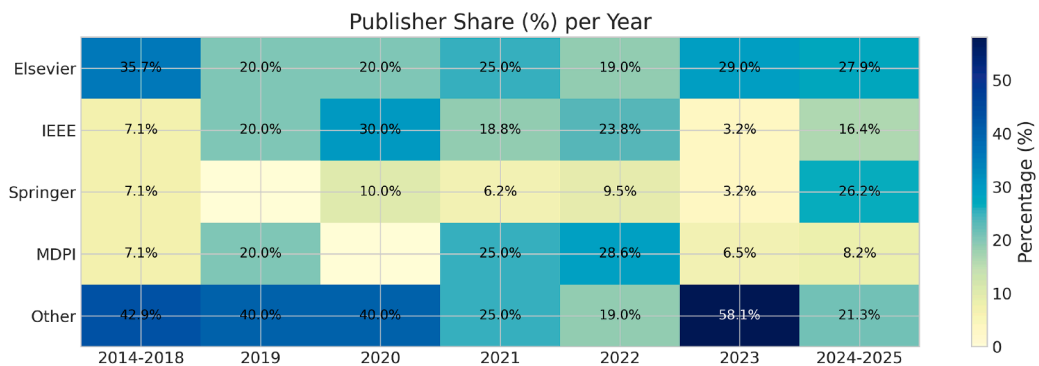


Fig. 5. Annual publisher share (%) of publications.

4. IoMT security

In connected healthcare environments, securing IoMT communications is one of paramount importance, given the sensitive nature of the transmitted medical data, the potential impact on patient safety, and the wide range of threats posed by the interconnectivity and heterogeneity of IoMT devices within complex healthcare networks [17]. The security of IoMT networks enhances trust in the data collected by sensors, enabling decisions to be made based on reliable and authentic data. This prevents unauthorized insertion or modification of data, thereby helping to protect and secure both devices and patients [18]. Among the objectives of IoMT security are

ensuring the availability of devices, systems, and services throughout their lifecycle [19]. This section aims to shed light on the key mechanisms of IoMT by describing its architecture, cybersecurity imperatives, identified risks, and recent technological advancements in the connected medical field.

4.1. IoMT architecture

IoMT architecture has the perception layer (Environment-aware layer), the network layer (Gateway layer), the cloud layer and the application layer (function layer) [20]. Fig. 6 presents the architecture of an IoMT system.

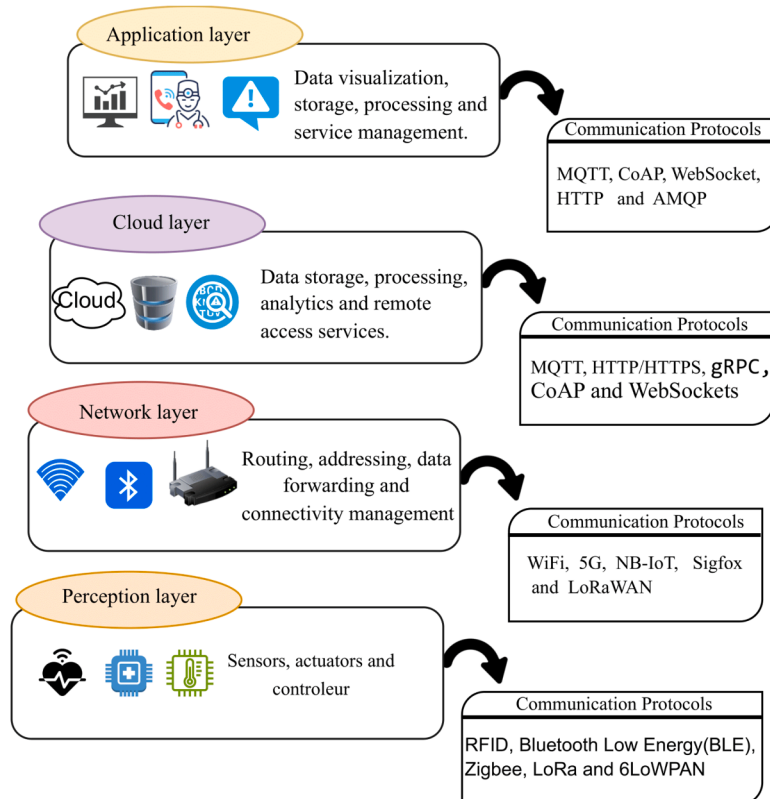


Fig. 6. Layered IoMT architecture with communication protocols.

4.1.1. The perception layers

This layer includes various medical devices, such as scanners, monitors, connected objects, and sensors designed to measure blood glucose levels, oxygen saturation, temperature, heart rate, blood pressure, respiratory rate, and electrocardiographic (ECG) activity. It also encompasses smart insulin pumps, connected implants, and health-tracking wristbands [21]. These devices capture sensitive medical information in realtime and relay it to the network layer for processing. Given the power constraints inherent to IoMT devices, data transfer is facilitated through short-range communication methods such as Bluetooth, Zigbee, Near Field Communication (NFC), and Radio Frequency Identification (RFID). For long-range, low-power communication, technologies like LoRa (Long Range) are also employed, enabling connectivity over several kilometers while maintaining energy efficiency [22,23]. The limited computing and storage capacity of IoMT devices often only allows them to perform the preprocessing stage, such as verifying correct data reception, checking data structure, removing duplicates, normalizing measured values, or filtering noisy data before transmitting it to more powerful processing systems.

4.1.2. The network layer

The network layer is responsible for transferring collected data, such as network traffic and patient biometric information, which are initially gathered by the perception or application layer. These data are then sent to the cloud or a cloud computing platform for analysis and storage. In some cases, intelligent processing is performed in the cloud to generate commands that are sent to connected devices. Among the commonly used communication protocols at this layer are 5G, Wi-Fi, Sigfox, NB-IoT, and LoRaWAN, especially for enabling data transmission in Low Power Wide Area Networks (LPWANs) [24].

4.1.3. The cloud layer

The cloud layer in IoMT is essential for the storage, access, and analysis of medical data. It enables real-time patient monitoring, thereby facilitating telemedicine and clinical decision-making. Cloud technologies also provide solutions for the secure management of healthcare data [25]. In this context, cloud services are divided into several models, Infrastructure as a Service (IaaS) provides the necessary hardware resources such as servers, storage, and networks; Platform as a Service (PaaS) offers a complete environment for developing, testing, and deploying medical applications; and Software as a Service (SaaS) allows access to ready-to-use medical software without the need to manage the underlying infrastructure [26].

4.1.4. The application layer

In the architecture of the IoMT, the application layer plays a crucial role by serving as the interface between connected medical devices, patients, healthcare professionals, and analytics systems. It enables real-time visualization of physiological data (such as heart rate, temperature, or blood glucose levels) through userfriendly interfaces like smartphones or web platforms. This layer also supports remote management of medical devices, continuous health monitoring, access to electronic medical records (EMR) [27], and predictive risk analysis. It facilitates teleconsultation, allowing doctors to diagnose, monitor, and advise patients remotely, thus improving access to healthcare, particularly in rural or underserved areas. Additionally, intelligent chatbots embedded within this layer enable seamless communication between patients and doctors both online and offline, providing immediate responses, preliminary consultations, and guidance, which enhances patient engagement and care continuity [28]. From a cybersecurity perspective, the application layer is the most frequently targeted by malicious attacks, primarily because it provides direct interfaces to a wide array of communication protocols, connected devices, and critical datasets that contain highly sensitive patient information as well as machine operational data. This layer acts as the gateway between users and the underlying infrastructure, making it an attractive target for adversaries aiming to exploit vulnerabilities for unauthorized access, data breaches, or system manipulation. Effective security measures at the application layer are therefore paramount, as this is the most exposed and externally accessible component—often reachable via the internet or other wide-area networks. Securing the application layer involves not only protecting the codebase by implementing rigorous software development lifecycle practices, such as secure coding, thorough testing, and timely patch management, but also enforcing strong authentication, authorization, and encryption mechanisms to minimize vulnerabilities. Failure to properly secure this layer can compromise the integrity, confidentiality, and availability of the entire healthcare system, potentially leading to catastrophic consequences for patient safety and data privacy [29]. Ensuring its security helps guarantee the protection of other layers in the IoMT architecture. Consequently, this layer integrates advanced network monitoring tools that help administrators detect abnormal behavior, block unauthorized access, and prevent the hacking of sensitive medical data [30]. Through the integration of dashboards, intelligent alerts, decision support systems, and chatbots, the application layer transforms collected data into actionable insights for effective, secure, and personalized medical follow-up [31].

4.2. Security requirements for IoMT systems

According to several research studies [18,20,32–34], the security of IoMT devices is primarily based on three core requirements: confidentiality, integrity, and availability, namely CIA.

✓ **Confidentiality:** it ensures the security of patient data collected by medical sensors, including treatment details and personal information such as National ID, phone number, or banking details. Furthermore, it prevents unauthorized access by third parties, whether during the storage or transmission of data generated by IoMT devices.

✓ **Integrity** in the medical field, as defined within the CIA triad, refers to the assurance that health information and data transmitted through the medical infrastructure remain accurate, reliable, and consistent, except in cases of duly authorized modifications. Any alteration may result from improper access, negligent use, technical errors, or unauthorized intrusions. Integrity is considered preserved as long as medical data remain unchanged during storage, transmission, and use, in the absence of legitimate modifications.

✓ **Availability** this means that information must be accessible when authorized users need it. In the medical field, each patient should be able to access their health data, just as the responsible physicians must be able to consult their patients' medical records. This information is generally displayed through web applications, such as dashboards, or mobile applications.

4.3. Threats to IoMT security

With the rapid expansion of IoMT, the adoption of complex technologies and layered communication infrastructures increases exposure to cybersecurity threats. The limited computational resources of IoMT devices, the heterogeneity of medical data, and the dynamic nature of their networks create a fertile ground for attacks, rising critical concerns about ensuring data confidentiality, integrity, and availability [8,35–39]. This requires a thorough analysis of devices, communication protocols, and user-side backend infrastructures in order to identify potential vulnerabilities across the entire IoMT ecosystem. Attack identification is based on the analysis of the existing architectural layers, as each layer handles particular data flows and exposes distinct vulnerabilities. Among the most common attack types in IoMT environments are Denial of Service (DoS), Distributed Denial of Service (DDoS), reconnaissance attacks, Man-in-the-Middle (MiTM), and spoofing attacks, each affecting one or more aspects of the CIA triad depending on the attack vector and target layer [33,40–42]. To gain a better understanding of how attacks manifest within the IoMT environment, it is essential to classify them according to the architectural layer they target [1,43,44]. The classification of attacks can be further refined by analyzing five key dimensions: the nature of the attack, its scope, the target, the attacker's capacity, and the impact, as illustrated in Fig. 7.

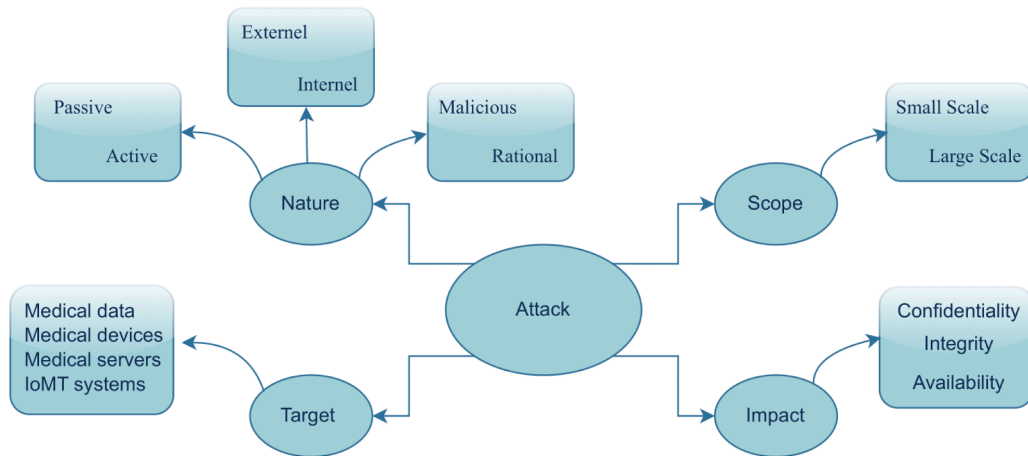


Fig. 7. Classification of attacks in healthcare systems: Targets, Scope, and Impact.

- **Attacker's Nature** refers to the fundamental characteristics and intentions of the threat actor. It encompasses four primary categories: passive vs. active attacks and internal vs. external attacks.
 - **Passive attacks** aim to eavesdrop or monitor data without altering system resources, thereby remaining undetected.
 - **Active attacks**, on the other hand, involve direct interaction with the system to modify, disrupt, or damage its operations, such as in DoS or injection attacks.
 - **Internal attacks** originate from legitimate users or compromised insiders with authorized access, making them particularly dangerous due to their knowledge of the system.
 - **External attacks** are carried out by unauthorized actors who attempt to penetrate the system from outside the network boundary.
- **Attack target:** refers to the fundamental characteristics and intentions of the threat actor. It encompasses four primary categories: passive vs. active attacks and internal vs. external attacks.
 - **Medical data** represents one of the most sensitive and valuable assets within the IoMT ecosystem. It includes EHRs, patient identifiers, diagnostic images, real-time monitoring data, and treatment histories. Due to its high value on the black market and its potential to cause serious harm if altered or exposed, medical data is a primary target for cyberattacks. Attackers may attempt to steal this data for identity theft or financial fraud, alter it to compromise patient safety, or encrypt it in ransomware attacks to demand payment from healthcare providers. Moreover, unauthorized access to such data violates patient privacy and can lead to severe legal and regulatory consequences. Ensuring the confidentiality, integrity, and availability of medical data is therefore a top priority, requiring robust encryption, secure data transmission protocols, strict access controls, and continuous monitoring for potential breaches [8,20].

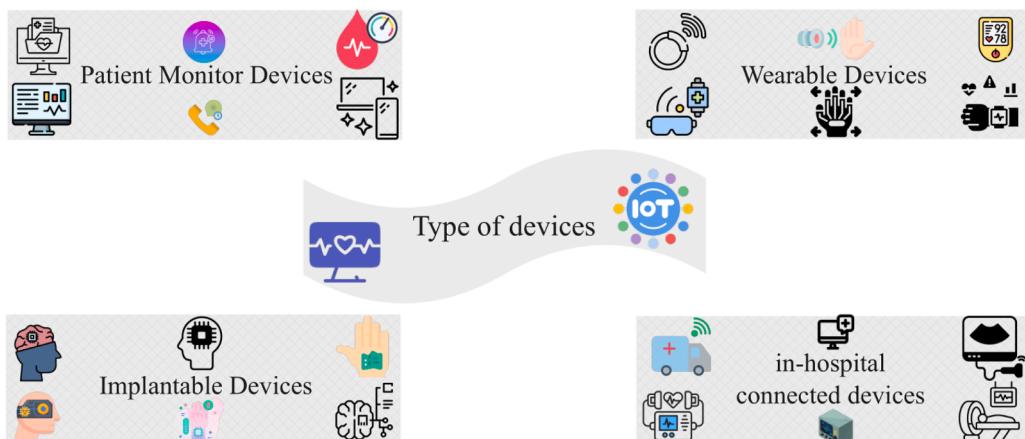


Fig. 8. Types of medical devices in IoMT ecosystems.

- **Medical devices** in the context of the IoMT, various medical devices are increasingly interconnected, enabling real-time patient monitoring and data exchange. However, this connectivity also introduces significant cybersecurity risks that can compromise

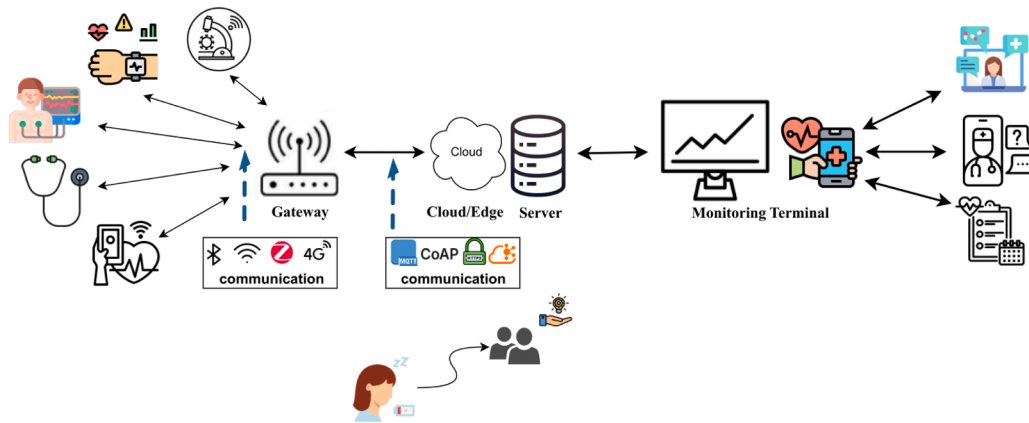


Fig. 9. Medical data transmission architecture: edge device to cloud platform.

both patient safety and data integrity. Fig. 8 highlights the various categories of medical devices vulnerable to cyberattacks. It shows that patient monitors such as cardiac monitors or pulse oximeters may suffer from data flow interruptions or result tampering. Implantable devices like pacemakers or insulin pumps are at risk of malicious parameter modifications. Wearable devices including smartwatches or medical patches are exposed to personal data theft or the injection of false diagnoses. Lastly, connected hospital equipment such as imaging systems or electronic health records can be targeted by ransomware or the interception of sensitive data [17,45]. Despite the increasing complexity of cyber threats, many medical devices suffer from inherent hardware limitations that make it difficult to implement robust security mechanisms. These limitations include constrained memory size, low computational power, and limited energy resources particularly in implantable and wearable device [46,47]. As a result, deploying conventional security algorithms such as encryption, intrusion detection, or anomaly detection models become a challenge. These constraints must be carefully considered when designing lightweight yet effective cybersecurity solutions for IoMT environments.

- **Medical servers** are a critical aspect of cybersecurity in the field of the IoMT. These servers centralize sensitive data and play a key role in the operation of healthcare infrastructures, which make them prime targets for cyber attackers.
- **IoMT systems** are composed of multiple interconnected components that work together to achieve a global objective: enabling data visualization and sharing among various stakeholders, including hospitals, doctors, and even the patients themselves. These systems typically involve a data flow starting from medical devices (such as wearable sensors or implantables), passing through gateways and medical servers either local or hosted on cloud computing infrastructures for data storage and processing. The information is then accessed through IoT platforms, which can be desktop applications, mobile apps, or both. However, this complex architecture also introduces multiple cybersecurity challenges. Each component whether it's the device, the network gateway, the server, or the user interface represent a potential attack surface. Common threats include data interception, unauthorized access, ransomware, tampering with medical records, DDoS and DoS attacks. Ensuring end-to-end security in IoMT systems is therefore crucial to protect patient data and maintain trust in digital healthcare environments. Fig. 9 illustrates a basic data transmission scheme within an IoMT system, highlighting the different layers involved in the communication process.

4.3.1. Attack at perception layer

The perception layer, serving as the interface between the physical world and the IoMT system, is primarily responsible for data acquisition through medical sensors and devices. Due to its proximity to the patient and exposure to uncontrolled environments, it is particularly vulnerable to various physical and low-level cyberattacks. Table 2 presents a compilation of different studies focusing on various perception layer technologies, such as WSN, RFID, LoRaWAN, and Bluetooth. Each technology is associated with specific types of attacks that impact one or more components of the CIA triad (Confidentiality, Integrity, and Availability).

4.3.2. Attack at network layer

The network layer of IoMT systems exhibits intrinsic vulnerabilities tied to its operation. Its reliance on wireless communication exposes medical data to risks of interception or manipulation by malicious actors. Additionally, the dynamic and open nature of these networks facilitates infiltrations, particularly through compromised nodes or targeted attacks. The criticality of real-time data exchanges amplifies these risks, as even brief disruptions can lead to detrimental delays in urgent medical contexts. Among the major threats are sinkhole attacks, where a malicious node diverts traffic to block or fraudulently analyze data. Selective forwarding allows compromised entities to discreetly drop specific packets, compromising the integrity of data flows. Sybil attacks, which rely on the spoofing of multiple identities, distort routing mechanisms or network consensus. Furthermore, wormhole attacks create covert tunnels to intercept, alter, or replay communications, while DoS attacks paralyze the network through artificial overload. These threats directly undermine the principles of CIA. Leaks of sensitive data, corruption of transmitted information, or service unavailability can

Table 2
Summary of attacks targeting IoMT perception layer technologies.

Technology	Attack	CIA	Ref	Year
WSN	Jamming Attack	Availability	[48]	2020
	Tampering Attack	Confidentiality, Integrity		
	Collision	Availability, Integrity		
	Exhaustion	Availability		
RFID	Relay Attack	Confidentiality, Integrity	[48]	2020
	Permanently Disabling Tags	Availability		
	Temporarily Disabling Tags			
WSN	Node Tampering	Confidentiality, Integrity	[8]	2023
	Sybil Attack	Availability, Integrity	[49]	2024
	<i>Battery Exhaustion</i>	Availability	[50]	2023
LoRaWAN	Eavesdropping Attack	Availability	[51]	2020
	Destroy/Remove End-device			
	Jamming Attack			
	ACK Spoofing Attack			
	False Join Packets			
	Downlink Routing Attack			
	Replay Attack			
Bluetooth	Bluesnarfing	Confidentiality	[52]	2018
	Bluejacking	Confidentiality, Integrity		
	Blueborne	Availability		
	BlueBugging			

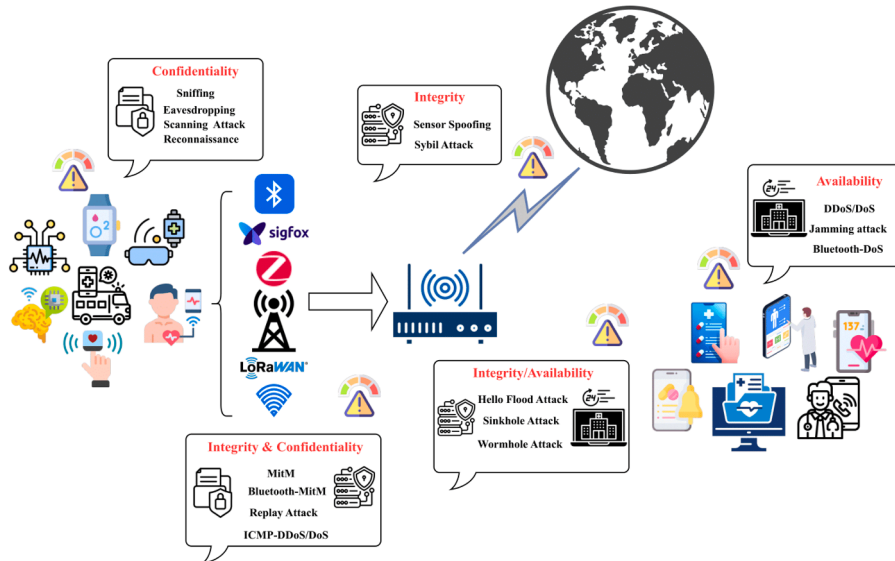


Fig. 10. Network layer attacks.

lead to diagnostic errors, inappropriate treatments, or life-threatening risks for patients.

Fig. 10 illustrates these various attacks and maps their respective impacts on the Confidentiality, Integrity, and Availability of IoMT systems, providing a visual summary of the security challenges that must be addressed. Securing this layer thus demands proactive strategies to counteract routing manipulations, unauthorized eavesdropping, and targeted disruptions, while preserving the smooth flow of communications essential to medical care [3,8,53].

4.3.3. Attack at cloud layer

The cloud is vulnerable to attacks such as phishing, data leaks, DoS/DDoS attacks, and malware, targeting both infrastructures and sensitive data [54]. Fig. 11 maps these cloud-layer attacks to their specific impacts on CIA, highlighting critical vulnerabilities in IoMT systems. The visual taxonomy underscores security challenges such as data breaches (compromising confidentiality), injection attacks (threatening integrity), and DDoS disruptions (undermining availability).

Effective protection requires proactive measures against insider threats, adversarial manipulations, and service disruptions ensuring uninterrupted, secure healthcare data flows.

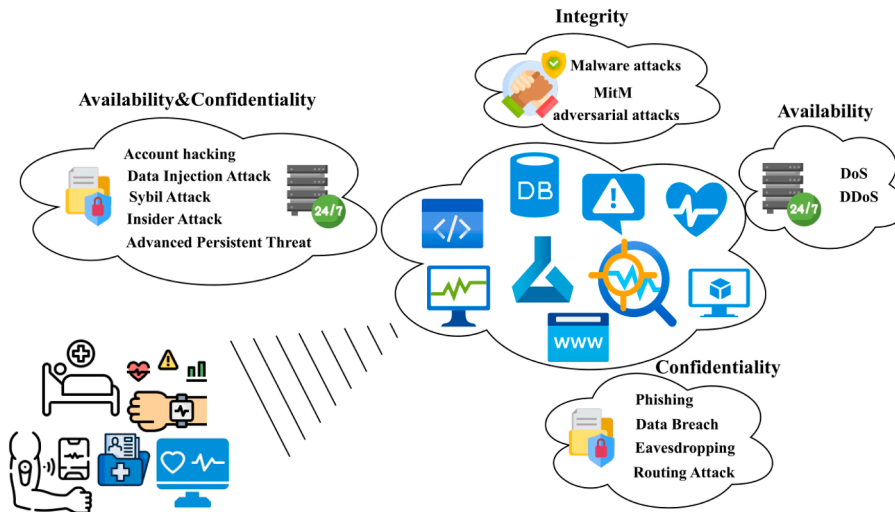


Fig. 11. IoT cloud security risks.

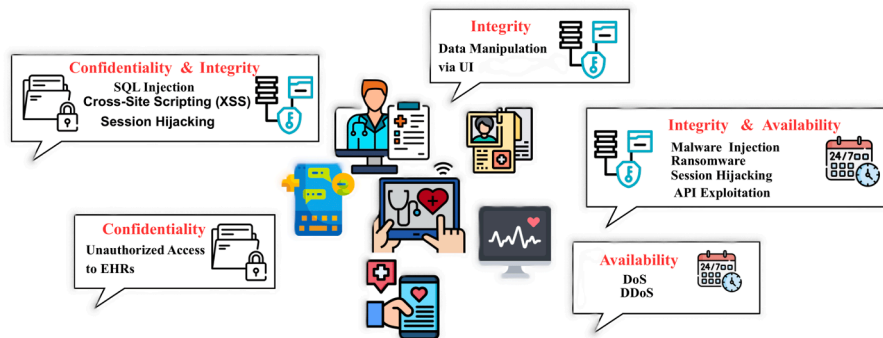


Fig. 12. Application layer security risks.

4.3.4. Attack at application layer

In IoMT systems, the application layer is particularly vulnerable to attacks targeting medical software, APIs, and mobile health applications. Threats such as malware injection, cross-site scripting (XSS), SQL injection, buffer overflow, session hijacking, unauthorized access to electronic health records (EHRs), and manipulation of data through insecure user interfaces can severely compromise patient safety, data integrity, and confidentiality [39]. Fig. 12 illustrates various application-layer attacks and highlights their potential impact on IoMT system security.

4.4. Most common cyberattacks in healthcare

This subsection classifies the most reported cyberattacks in recent healthcare cybersecurity literature into eight main categories: Ransomware attacks, DDoS attacks, AI-powered phishing, Social Engineering, DoS attacks, MitM attacks, Deepfake-based attacks, and Malware attacks. Each category highlights the targeted IoMT layer and the specific security aspects of the CIA triad that are compromised.

4.4.1. Ransomware attacks

Ransomware attacks aim to deny users access to critical system files by encrypting them and demanding a ransom payment for their release [55]. This form of cyberattack poses a serious and growing threat to hospitals, primarily due to the substantial financial costs involved and the significant disruptions it causes to healthcare services. Consequently, hospitals are increasingly concerned about ransomware incidents, given their profound economic impact and the severe interruptions they inflict on the delivery of essential medical care [56]. In 2018, Indiana hospital pays 55,000 after ransomware attack [57]. In September 2020, a ransomware attack targeted the University Hospital of Düsseldorf in Germany, disrupting system and data access. This disruption compelled the hospital to redirect emergency patients to other facilities. Unfortunately, a woman who was transferred to a hospital about 30 km away succumbed due to the delay in receiving urgent medical care [58]. In December 2022, hackers attempted a ransomware attack on a Louisiana (USA) hospital system, compromising sensitive data of before being disrupted. The breach exposed personal and medical

records, though the attackers' ransom demands were not met. Authorities investigated potential ties to known cybercriminal groups [59]. In May 2025, a large-scale ransomware attack targeted several hospitals and clinics across the United States, crippling their computer systems and disrupting patient care. The hackers demanded a cryptocurrency ransom to restore access to the data. U.S. authorities, including the FBI, were mobilized, suspecting the involvement of a Russian cybercriminal group or an entity affiliated with a hostile state [60].

4.4.2. Malware attacks

Malware attacks represent a significant cybersecurity threat to medical institutions, as they involve malicious software designed to infiltrate, disrupt, or damage hospital information systems and compromise sensitive patient data [61,62]. The high value of personal health data on illicit markets makes medical institutions prime targets for malicious software specifically designed to steal or encrypt (ransomware) this sensitive information [63]. In 2019, Michael Garron Hospital fell victim to the Ryuk ransomware, which spreads via phishing and remains dormant while gathering information before encrypting files and demanding ransom. As a result, the hospital had to revert to manual operations, and several patient appointments were canceled or rescheduled. Similar disruptions were reported in other hospitals, including in Alabama and Australia. Within five months, Ryuk is estimated to have generated approximately 3.7 million for its operators [64].

4.4.3. AI-powered phishing

With the rise of AI, phishing attacks have become increasingly sophisticated. AI-powered phishing leverages ML algorithms and natural language processing to craft highly personalized, convincing messages that are harder to detect. In the healthcare sector, such attacks pose a serious threat, as they can deceive medical staff into revealing sensitive credentials or accessing malicious links, potentially compromising entire hospital systems [65]. Several types of AI-powered phishing have emerged. For example, [66] presents six forms: Popup phishing (fake alerts or login windows), Watering hole attacks (infecting frequently visited websites), Phishing-as-a-service (ready-to-use AI phishing kits), Vishing (voice phishing using AI-generated voices), Ransomware via phishing (AI-personalized emails delivering malware), and Angler phishing (impersonating support accounts on social media to deceive users).

4.4.4. Advanced persistent threats (APTs)

APTs are sophisticated and long-term cyberattacks, usually conducted by organized groups, that discreetly target IoMT systems to steal sensitive data or cause operational disruptions [67].

4.4.5. Deepfakes

Deepfakes are synthetic media generated using DL techniques, which can convincingly mimic real human voices, faces, or actions, posing serious threats to identity verification, misinformation, and trust in digital communications [68]. In the healthcare sector, deepfakes are emerging as a major cyberattack type, particularly through audio formats, which are harder to detect than video deepfakes. These manipulated recordings can be used to steal personal credentials or trick victims into triggering malicious software, potentially disrupting entire computer networks. Voice deepfakes, with a projected annual global growth rate of 37.6% from 2024 to 2033, represent a growing threat vector that requires increased vigilance and awareness [69].

4.4.6. DDoS attacks

DDoS attacks are a type of cyberattack in which multiple compromised devices are used to overwhelm a target system, server, or network with excessive traffic, rendering it unavailable to legitimate users [70]. In the context of healthcare and IoMT, such attacks can severely disrupt critical medical services and compromise patient safety by causing system outages and delays in care [71]. For example, in November 2023, a prolonged DDoS attack disabled the websites and Email systems of the Singapore public hospital network for approximately seven hours, disrupting internet services in 46 institutions before connectivity was restored [72]. In 2014, Boston Children's Hospital was targeted by an Anonymous hacktivist DDoS campaign that unexpectedly affected its network for about a week, halting online scheduling and communications and resulting in over 300,000 in lost donations [73]. More recently, the pro-Russian group KillNet launched DDoS waves against Dutch hospitals in early 2023, leading to acute care disruptions and cancelled operations [74]. In France, the Nantes University Hospital suffered a DDoS attack during the night of January 14, to 15, 2024, resulting in a complete outage of internet access, appointment booking tools, and Email services. However, internal medical software remained operational, illustrating the potential impact of such attacks on healthcare services [75].

4.4.7. DoS attacks

DoS attacks are malicious attempts to disrupt the normal functioning of a targeted system or service by overwhelming it with a flood of illegitimate requests, making it inaccessible to legitimate users. According to Palo Alto Networks, "A denial-of-service (DoS) attack is a cyber attack that inundates a system, application, or network with excessive traffic or resource requests, rendering it unavailable to legitimate users and disrupting operations" [76].

4.4.8. MitM attacks

MitM attacks occur when an attacker secretly intercepts and potentially alters the communication between two parties, enabling theft of sensitive data, unauthorized access, and disruption of normal communication posing significant risks to the security and privacy of healthcare and IoMT systems [3,77].

4.4.9. Supply chain

In 2025, healthcare supply chains are expected to become a prime target for cybercriminals, who increasingly exploit vulnerabilities in smaller third-party vendors to gain access to larger healthcare institutions and steal sensitive patient data. This highlights the critical need for healthcare organizations to rigorously assess the cybersecurity posture of their suppliers, ensuring adequate encryption, multi-factor authentication, and strong password policies to mitigate supply chain risks [69].

4.5. Real cyberattacks impacting hospitals

In 2021, the ransomware attack on Hillel Yaffe Medical Center demonstrated the devastating impact a sophisticated cyberattack can have on a fully digitized hospital infrastructure. By leveraging advanced encryption methods and legitimate system tools like Microsoft BitLocker, the attackers were able to bypass existing security protections and paralyze the hospital's entire digital ecosystem. The complete shutdown of access to EMR, laboratory results, imaging, patient scheduling, and internal communications severely disrupted both inpatient and outpatient care. This incident highlights the critical vulnerability of healthcare institutions to cyber threats and underscores the urgent need for robust cybersecurity strategies, incident response plans, and resilient system architectures to protect patient data and ensure the continuity of medical services [56]. In Abbou et al. [78], the authors conducted a retrospective analysis of six cyberattacks targeting public hospitals in the Portuguese National Health System between 2017 and 2022. Using simulation models, the study estimated that a single attack could result in direct financial losses ranging from €115,882.96 to €2,317,659.11, depending on the hospital size and duration of service disruption. These attacks not only caused delays in clinical activities such as outpatient consultations, emergency care, and surgeries but also posed serious operational and economic challenges. The findings highlight the vulnerability of healthcare services to cyber threats and the urgent need for investment in cybersecurity planning, staff training, and infrastructure resilience. In September 2024, hackers breached the Texas Tech Health Sciences Center's systems in Lubbock and El Paso, compromising the sensitive personal and medical data of over 1.4 million patients including names, dates of birth, Social Security numbers, driver's licenses, insurance and billing information marking yet another significant instance of cyberattacks exploiting healthcare institutions' vast data troves [79]. A recent study [80], analyzed cybersecurity data breaches across rural and urban hospitals in the United States over the period 2016-2022. Using a two-stage statistical approach, the researchers found that 3.46% of hospitals experienced a data breach, with urban hospitals being slightly more affected (3.84%) than rural ones (2.64%). The most vulnerable institutions were larger, nonprofit, and non-affiliated hospitals, especially those located in the Northeastern U.S. Interestingly, once a breach occurred, no significant differences were found between rural and urban hospitals in terms of breach type or location hacking and IT-related incidents were dominant in both contexts. The study also revealed an increasing trend in attacks targeting EMR and network servers, reflecting the accelerating digitalization of healthcare services and the evolving tactics of cybercriminals. These results highlight the need for targeted, resource-aware cybersecurity strategies, including improved threat detection, workforce training, and incident response planning. As hospitals continue to digitize, this recent research provides valuable insights for policymakers, health system leaders, and cybersecurity professionals seeking to strengthen data protection and care continuity in an increasingly connected environment. To raise awareness about the severity of cyberattacks that can potentially lead to the shutdown of entire hospitals, the Assistance Publique-Hôpitaux de Paris (AP-HP) has taken a proactive approach by conducting realistic ransomware simulation exercises, known as the 'CRYPTOlocker' exercises, since May 2021. These simulations aim to strengthen emergency preparedness, response coordination, and cyber-resilience. The most recent full-scale exercise, held on July 5, 2023, mobilized over 200 participants across the hospital network to simulate the real-world consequences of a ransomware attack and improve institutional readiness against such critical threats [81].

4.6. Intrusion detection systems (IDS)

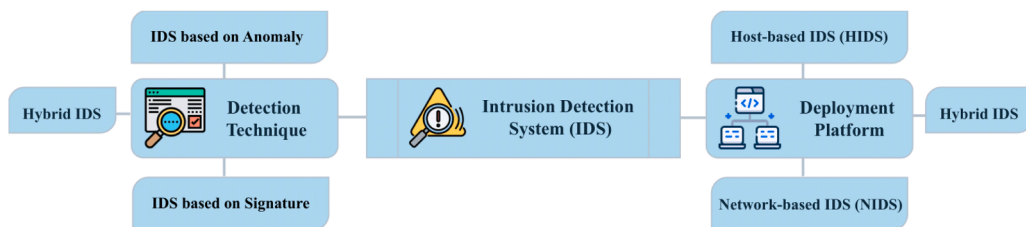


Fig. 13. Taxonomy of IDS.

An IDS continuously and autonomously monitors network traffic and notifies administrators when it detects abnormal or potentially malicious behavior [82,83]. In the context of the IoMT, IDSs are essential to detect cyberattacks by analyzing patterns based on signatures, anomalies, or behaviors, thereby protecting sensitive medical data and ensuring the availability of critical healthcare services. In general, IDSs act as the first line of defense in identifying and responding to security threats targeting IoMT environments [1]. IDS can be classified into signature-based and anomaly-based detection methods [56], as shown in Fig. 13. Deployment platforms for IDS are classified into three types: Network-based IDS (NIDS), Host-based IDS (HIDS), and Hybrid IDS. The HIDS is designed to monitor and assess the internal mechanisms and activities of a computing system in real-time. The NIDS functions within a network

Table 3
Overview of open-source HIDS and NIDS tools for multiple operating.

Software	Type	Description
OSSEC [84]	HIDS	Open-source HIDS for Windows, Linux, macOS. Monitors logs, detects rootkits, checks file integrity, and offers centralized management. Known for active response capabilities.
Samhain [85]	HIDS	Lightweight HIDS with Linux, macOS, Windows (via Cygwin) support. Focuses on file integrity, log monitoring, stealth detection, and encrypted client-server communication. Less actively maintained.
Wazuh [86]	HIDS	Advanced HIDS based on OSSEC; provides real-time threat detection, file integrity monitoring, log analysis, vulnerability detection, and integration with SIEM platforms.
Suricata [87]	HIDS	Multi-threaded IDS/IPS capable of deep packet inspection, flow analysis, and file extraction; supports intrusion detection at network level and basic hostlevel features.
Zeek [88]	NIDS	Network monitoring tool focused on behavioral analysis and traffic logging; ideal for identifying anomalies and creating custom detection policies.
Snort [89]	NIDS	Signature-based NIDS and IPS for real-time packet inspection and traffic analysis; detects known attacks such as buffer overflows, port scans, and DoS attempts.

environment where packet capture and analysis are performed to facilitate the application of appropriate security measures. A hybrid IDS combines signature-based and anomaly-based approaches, thereby providing more accurate and comprehensive detection of network threats. Table 3 presents various open-source HIDS and NIDS software available on the market, supported by Linux, Windows, and macOS operating systems.

5. Artificial intelligence algorithms for IDS

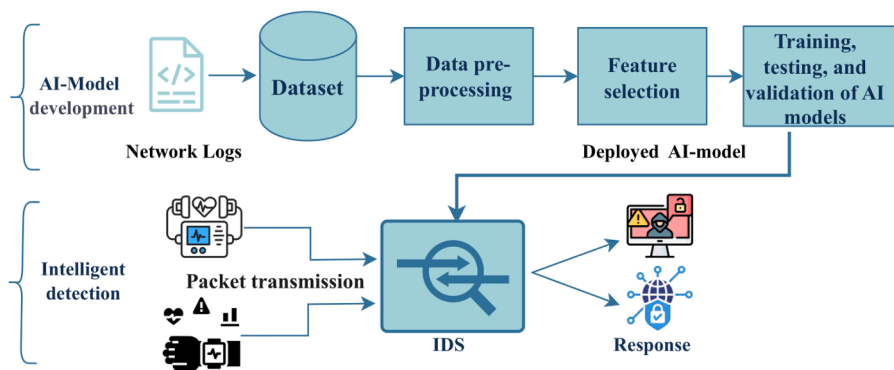


Fig. 14. AI-powered IDS for Healthcare: From data processing to threat response.

AI plays a fundamental role today in the field of cybersecurity by providing innovative and adaptive solutions in response to the growing complexity of digital threats. Unlike traditional approaches, which are often based on static signatures or fixed rules, AI enables dynamic, continuous, and intelligent analysis of network and system behaviors. It is particularly effective at detecting unknown zero-day attacks, subtle anomalies, or malicious behaviors hidden within legitimate traffic [90]. Among the most prominent applications of AI in cybersecurity, IDS stand out as extensively explored and practically impactful solutions. These systems aim to identify unauthorized access attempts or malicious activities on a network, and they greatly benefit from the learning capabilities of AI to better distinguish between normal and suspicious behaviors. Fig. 14 illustrates the workflow of an AI-based IDS system. The development of such a system starts with the collection of network logs, which are then used to construct a dataset. This dataset undergoes pre-processing and feature selection, ensuring that the most relevant information is retained for training, testing, and validating AI models. Once trained, the model is integrated into the IDS and operates in real time, analyzing incoming traffic to detect abnormal behaviors. When a threat is detected, the system can automatically initiate appropriate response actions. This intelligent detection mechanism greatly improves both the accuracy and speed of threat identification, making AI-based IDS particularly suitable for protecting complex and sensitive environments like the IoMT. As illustrated in Fig. 15, AI encompasses various subfields such as ML and DL, which provide the technical foundation for many modern cybersecurity solutions. Through ML and DL, AI can learn from large datasets, recognize complex patterns, adapt to evolving threats, and even anticipate certain attacks before they occur. By integrating these algorithms into cybersecurity tools, organizations observe significant improvements in detection precision, reduced false positives, and faster incident responses [91]. Finally, Fig. 16 provides a structured overview of AI-based algorithms, categorizing them into ML and DL approaches, each further divided into supervised and unsupervised techniques. This classification reflects the

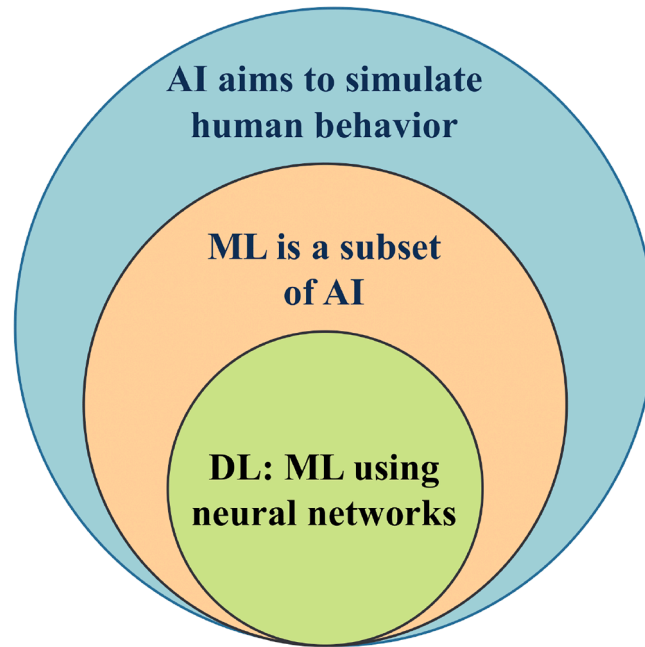


Fig. 15. The AI subset hierarchy: ML and DL.

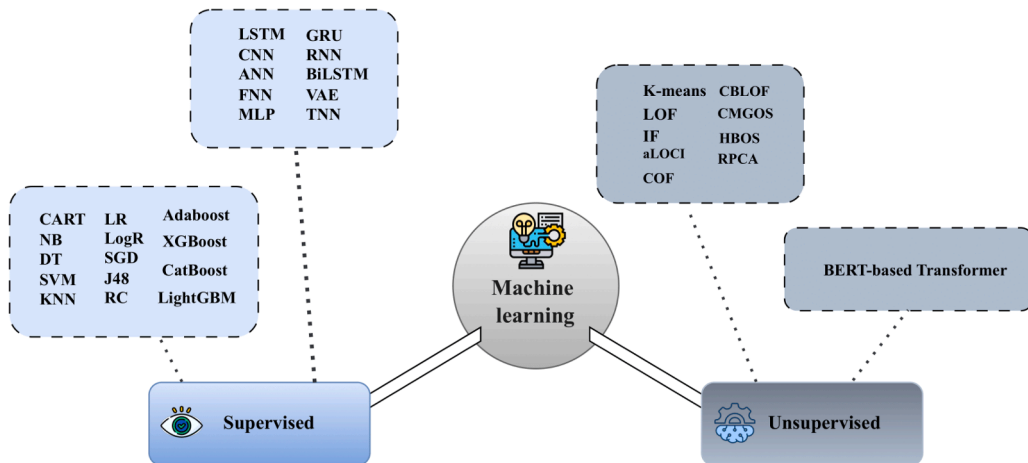


Fig. 16. AI algorithm hierarchy for IoMT solutions.

diversity of models such as DT, neural networks, clustering methods, and transformers used to enhance intelligent systems across domains, including IoMT and cybersecurity.

5.1. AI framework for IDS: from data preprocessing to evaluation metrics

5.1.1. Data preprocessing

Data preprocessing is a fundamental step in the development of AI-based IDS, as it ensures the quality and consistency of the input data. This phase includes operations such as data cleaning, normalization, encoding of categorical variables, and handling of missing values, which collectively enhance the performance and reliability of ML models [92].

5.1.1.1. Techniques for Handling Missing Data. Missing values are a common issue in intrusion detection datasets and can compromise both the learning process and model performance. These missing entries often result from incomplete data acquisition or system faults. If not handled appropriately, they can distort the distribution of features, introduce analytical bias, and negatively affect the predictive accuracy of ML algorithms [93]. There are several techniques for handling missing values. The simplest approach, known as listwise deletion, consists of removing entire rows that contain any missing entries a method frequently employed in many research studies

[94]. In some cases, researchers may also choose to eliminate entire columns if all the values in that column are missing. Other techniques also exist, such as mean substitution, where the missing value is replaced with the mean value of the same variable [92]. Other techniques also exist, such as mean substitution, where the missing value is replaced with the mean value of the same variable. Another commonly used method is Last Observation Carried Forward (LOCF), which replaces the missing value with the last previously observed value for that variable. This approach is particularly useful in time series or longitudinal data, as it helps maintain continuity in observations. However, it may introduce bias if the variable changes significantly over time. Other advanced methods for handling missing data include Multiple Imputation by Chained Equations (MICE) [95] and Maximum Likelihood Estimation (MLE) [94].

5.1.1.2. Data encoding. In ML and IDS systems, raw categorical data must often be converted into a numerical format to be properly interpreted by algorithms. This process, known as data encoding, includes several widely used techniques depending on the type of categorical features and the modeling context. These include label encoding, which is simple and effective for ordinal variables; one-hot encoding, which prevents artificial ordinal relationships by creating separate binary columns for each category; binary encoding, which is memory-efficient for features with many categories; target encoding, which replaces categories with the mean of the target variable to potentially improve supervised learning performance; frequency encoding, which captures the distribution of each category within the dataset; dummy encoding, a variant of one-hot encoding that avoids multicollinearity by dropping one category; hash encoding, which maps categories into fixed-length vectors and is particularly useful for high-cardinality data; and effect encoding, which expresses each category in terms of its deviation from the global mean and is especially suited for linear models. Each technique serves specific goals related to model performance, interpretability, or computational efficiency. As discussed in Dahouda and Joe [96] these eight encoding methods are among the most commonly applied in ML preprocessing.

5.1.1.3. Feature scaling. Feature scaling refers to the process of transforming the input features in a dataset so that they share a common scale or statistical distribution. It is considered a mapping technique applied during the preprocessing phase, where the goal is to ensure that all features contribute equally by giving them the same relative weight in the learning process. This step becomes essential when the dataset includes variables with widely different ranges or units, which can otherwise lead to models learning biased patterns [97]. In Cao et al. [98], the authors show that the choice of feature scaling techniques can significantly influence the overall performance of the model. They propose a novel method called the Generalized Logistic (GL) algorithm, which scales data by learning a generalized logistic function that fits the empirical cumulative distribution of the data. Unlike traditional methods such as Min-Max or Z-score scaling, the GL algorithm performs nonlinear scaling, is inherently robust to outliers, and is especially effective in clinical or biomedical contexts where datasets are typically small and noisy. The authors validate their approach on 16 binary classification tasks, showing that models trained on GL-scaled data achieve better performance in terms of AUROC and classification accuracy, outperforming standard scaling techniques. This work highlights the often-underestimated impact of preprocessing and introduces a simple yet powerful tool for improving model reliability in medical applications. A most recent study [97] provides a comprehensive evaluation of 12 feature scaling techniques, including both commonly used and less conventional ones. It presents real-world testing across multiple datasets and ML models, highlighting how different scaling methods can impact model performance depending on the context. Together, these studies emphasize the critical role of data preprocessing particularly feature scaling in building accurate and robust ML models.

5.1.1.4. class imbalance. Class imbalance is a major issue in ML that significantly affects classification tasks [99]. It can greatly influence the detection of attacks, as minority classes such as specific types of cyberattacks may be underrepresented in the dataset. This imbalance leads models to become biased toward the majority class, making it difficult to accurately detect rare but critical attacks [100]. The SMOTE (Synthetic Minority Oversampling Technique) technique is the most widely used and trending method for addressing the imbalanced data problem [101]. Instead of simply duplicating samples from the minority class, SMOTE generates new, synthetic examples by interpolating between existing minority samples and their nearest neighbors. This approach helps the model to better generalize by expanding the decision space of the minority class, thereby improving recall and overall classification performance. However, SMOTE may generate noisy or overlapping data if applied without caution, especially in high-dimensional feature spaces. Several improved variants such as Borderline-SMOTE, ADASYN, and SMOTE-ENN have been proposed to mitigate these limitations [100]. In Chen et al. [102], the authors present a comprehensive survey of the challenges posed by imbalanced learning, highlighting its impact on both ML and DL models in real-world applications. The paper not only reviews traditional classification techniques but also extends the analysis to regression tasks and deep long-tail learning. It offers a novel classification of existing methods and emphasizes the importance of practical applications across various domains, from management science to engineering.

5.1.2. Feature selection

Feature selection techniques play a crucial role in ML and pattern recognition by reducing computational cost, enhancing prediction performance, and improving the interpretability of models through a better understanding of the most relevant data features [103]. Feature selection techniques are generally categorized into three main approaches: filter methods [104], wrapper methods [105], and embedded methods. Each of these groups contains techniques that can be applied depending on the type of learning algorithm chosen whether supervised, unsupervised, or semi-supervised [106].

Table 4
Summary of evaluation metrics for classification models.

Metric	Equation	Components
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	TP: True Positives
Precision	$\frac{TP}{TP + FP}$	TN: True Negatives
Recall	$\frac{TP}{TP + FN}$	FP: False Positives
F1-score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	FN: False Negatives
Matthew's correlation coefficient (MCC)	$\frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$	

5.1.3. Training, validation, and testing in AI

The development of an AI model typically follows three critical phases: training, validation, and testing [107]. During the training phase, the model learns patterns from labeled data by optimizing its internal parameters. The validation phase is used to fine-tune the model's hyperparameters and monitor its ability to generalize, helping to avoid issues like overfitting. Finally, the testing phase evaluates the model's performance on completely unseen data, providing an unbiased estimate of how it will behave in real-world scenarios. Together, these phases form the foundation for building robust, reliable, and generalizable AI systems. Most articles use a 70% training, 20% testing, and 10% validation split [108]. However, some researchers only use training and testing phases without including a separate validation phase.

5.1.4. Evaluation metrics

The section on Evaluation Metrics presents the main performance indicators used to interpret and compare the results obtained by classification models in AI-driven cybersecurity research. These metrics are crucial for measuring the effectiveness of intrusion detection solutions, with particular attention to minimizing False Positives and False Negatives in IoMT environments. Table 4 summarizes these metrics along with their mathematical definitions and core components.

5.2. IDS based on ML models in IoMT

A branch of AI called ML processes information and generates predictions by applying data science, data mining, statistical methods, and analytical techniques. Training and testing are the two primary phases through which ML algorithms typically operate. ML is generally categorized into supervised learning, where models learn from labeled data, unsupervised learning, which identifies hidden patterns in unlabeled data and ensemble learning, which combines multiple models to improve prediction accuracy and robustness [109]. Table 5 presents studies related to the application of ML for IDS in the IoMT.

Tanzila Saba [110] proposed an IDS based on ensemble classifiers to secure IoMT devices in smart hospitals. This approach uses Principal Component Analysis (PCA) for feature reduction and several ensemble algorithms, including Bagged Decision Trees (DT), Random Forest (RF), Extra Trees (Bagging), AdaBoost, and Stochastic Gradient Boosting (Boosting), as well as a Voting Ensemble combining Logistic Regression (LR), CART, and SVM. The authors evaluated the model on the KDDCup-99 dataset. The results show that Bagged DT achieved the best performance, with an accuracy of 99.9%, precision of 93.2%, recall of 92%, and F1-measure of 96.1%.

Yao et al. [111] developed an edge-assisted anomaly detection system (EAD) to enhance the security of e-health systems. Their approach integrates feature selection optimized by BPSO with an ensemble learning model combining Bagging and Random Subspace methods, enabling the detection of four types of attacks: (1) attackers limiting user activity (e.g., Generic, Fuzzers, Exploits, Worms), (2) DoS/DDoS attackers flooding systems with excessive requests, (3) probe attackers collecting network information to exploit vulnerabilities, and (4) unauthorized access attempts involving packet injection or password guessing. Evaluated on the NSLKDD and UNSW-NB15 datasets, EAD achieved an accuracy of 90.06% and 90.59%, and a detection rate of 85.42% and 96.43%, respectively.

Kumar et al. [54] proposes a cyberattack detection framework for IoMT networks based on ensemble learning and a distributed fog-cloud architecture. This approach addresses the limitations of traditional cloud based security systems in the face of rising cyber threats, such as the 2018 ransomware attack on an Indiana hospital. The proposed system, named E-ADS, consists of two key components: a traffic processing engine that includes categorical feature mapping, correlation-based feature selection, min-max normalization, and an ensemble model combining DT, Naive Bayes (NB), and RF classifiers optimized using XGBoost, and an intrusion detection engine evaluated on the realistic ToN-IoT dataset, which better reflects modern IoMT threats than traditional datasets like NSL-KDD. Experimental results demonstrate that the proposed model outperforms state-of-the-art techniques in terms of accuracy, detection rate, low false alarm rate, and F1-score. Additionally, a deployment framework is introduced, leveraging a Software-as-a-Service (SaaS) model on the fog side and Infrastructure-as-a-Service (IaaS) on the cloud side. Future work aims to enhance feature optimization techniques and develop a functional prototype of the system.

Table 5
ML Model Classifications for IoMT Applications.

Year	Ref	ML Algorithm	Normalization	Feature Selection	Best Clf	Class Type	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Dataset
2020	[110]	Bagged DT, RF, AdaBoost, SGB, Logistic, CART, SVM	–	PCA	Bagged DT	Binary	93.2	99.9	92	96.1	KDD-99
2021	[111]	EDA, RF, SVM, AB, KNN, GNB, MLP	Min-Max	BPSO	EDA	Binary	90.06 90.59	– –	85.42 96.43	– –	NSL-KDD NSW-NB15
2021	[54]	DT, NB, RF, Ensemble learning	Min-Max	–	Ensemble learning	Binary	96.352	90.545	99.98	95.03	ToN-IOT
2021	[112]	RF, LR, NB	Min-Max	GA	GA-RF	Binary Multiclass	99.10 –	99.09 89.58	98.81 82.96	98.45 85.2	NSL-KDD CSE-CIC-IDS2018
2021	[113]	SVM, DT, RF, NB, LR, KNN	Min-Max	–	KNN	Binary	99.98	99.95	99.97	99.96	ToN-IOT
2021	[114]	RF, DT, KNN, RC, RNN	–	PSO	PSO-RF	Binary	99.76	99.75	96.45	–	NSL-KDD
2022	[115]	XGBoost, CatBoost, LightGBM	–	–	LightGBM	Binary	99.28	99.46	94.79	99.07	WUSTL EHMS 2020
2022	[116]	SVM	–	Pearson correlation	SVM	Binary	99.3	99.1	–	99.25	KDD-99
2022	[58]	LightGBM, BERT, BiLSTM	–	–	LightGBM	Binary Multiclass Binary	99.99 100 100 97.96	99.98 100 92 98.39	98.99 100 88 97.52	98.99 100 89 97.95	ToN-IOT Edge_IIoTset Edge_IIoTset EEMBER
2022	[117]	NB, KNN, DT, SVM, ANN	Z-score	–	DT	Binary	100	100	100	100	Bot-IoT
2022	[118]	RF, LR, DT, ET	Standard Scaler, Min-Max, Robust Scaler	–	RF	Binary	94.23	93.45	93.72	93.8	WUSTL EHMS
2022	[119]	LR, DT, SVM, RF, NB, IF, KM, LOF, DNN	Z-score	RFE, Chi-2	RF DT	Binary Multiclass	99.15 97.47	99 98.01	98.6 96.71	99.6 97.73	BlueTack
2023	[120]	DT, RF, KNN, MLP, NB	Min-Max	correlation matrix	DT(K=8)	Binary Multiclass	87.07 76.62	87.87 80.18	87.07 76.62	87.47 78.36	UNSW-NB15
2023	[121]	SVM, NB, GA-RF, Bayes Net, J48, RF	–	GA	GA-RF	Binary Multiclass	99.99 99.99	99.99 100	100 100	99.99 100	NSL-KDD UNSW-2018-IoT-Botnet
2024	[33]	RF, LR, DNN, AdaBoost	–	–	AdaBoost RF RF	Binary Multiclass 19 Type	99.6 73.5 73.3	95.9 73.5 69.1	96.1 71.3 57.7	95.9 67.6 55.1	CICIOMT2024
2024	[122]	RF, LR, DT, AdaBoost	Nilsimsa fingerprint	–	AdaBoost RF	Binary	98.37 99.48	99.17 100	98.24 77.51	98.70 98.72	CICIOMT2024 ICE
2024	[123]	DT, LogR, NB, RF, SVM	Z-score	–	RF DT RF	Binary (IP-based packets) Multiclass (IP-based packets) Binary (IP-based flows) Multiclass (IP-based flows)	99.87 99.47 99.90 99.52	99.87 99.49 99.90 99.52	99.87 99.47 99.90 99.52	99.87 99.48 99.90 99.52	IoMT-TrafficData
2025	[124]	LR, NB, AdaBoost, Stacking	Z-score	RFE	Stacking	Binary Multiclass	99 95.59	98 95.58	99 95.59	99 95.56	ECU-IoHT
2025	[125]	RF, XGB, DT, SVM	–	RFE, LFS, MI	RF	Multiclass Multiclass Multiclass	99 99 99.8	97.52 – 98.97	91.05 – 95.97	93.76 – 97.81	CICIOMT2024 CICIDS2017 NSL-KDD
2025	[126]	RF, XGB, DT, ADB, KNN	–	–	XGB	Binary	99.67	99.72	99.51	99.62	Healthcare-IoT dataset
2025	[127]	RF, CatBoost, LightGBM, XGBoost	–	–	RF	Multiclass	99.87	95	96.26	95.59	CICIOMT2024
2025	[128]	KNN, LSTM	Min-Max	LSTM-PCA	KNN	Binary Multiclass Multiclass Multiclass	99.9 99.1 99.8 98.2	99.9 98.8 99.7 86.2	99.9 98.7 99.7 96.5	99.9 98.7 99.7 89.3	WUSTL-EHMS ICU Ton-IoT ECU

Iwendi et al. [112] propose a Security of Things (SoT)-based intrusion detection framework tailored for smart healthcare systems, which are increasingly vulnerable to sophisticated web-based attacks and privacy violations. The study addresses the limitations of existing IDS approaches by introducing a ML-based support system that combines RF with a Genetic Algorithm (GA) for feature selection and optimization. The proposed model improves intrusion detection performance by selecting the most relevant features, normalizing data using the min-max technique, and transforming non-numeric attributes. Using the NSL-KDD dataset, the authors compared RF, NB, and LR classifiers, with RF outperforming the other models in terms of accuracy, detection rate, and false alarm rate (FAR). The optimized GA-RF model achieved a detection rate of 98.81% and FAR of 0.8%. Additionally, performance was evaluated using CSE-CIC-IDS2018 and different scaling factors, where the proposed method demonstrated significant gains in precision (+5.65%) and F1-score (+8.2%). DL models (CNN, RNN) were also explored for future improvements, along with the use of SMOTE to handle class imbalance in underrepresented attacks. The study concludes by proposing a secure communication framework for smart healthcare and highlighting plans to develop self-learning systems for real-time anomaly detection.

Zachos et al. [113] propose an anomaly-based intrusion detection system (AIDS) specifically designed for IoMT networks, which are vulnerable due to their heterogeneity and limited resources. The system employs both host-based and network-based approaches to monitor log files from IoMT devices and network traffic from the IoMT gateway, while taking computational constraints into account. The detection engine is deployed at the gateway level using a Raspberry Pi, and relies on lightweight ML algorithms to identify abnormal behavior. The authors evaluated their system using two datasets: the ToN-IoT Telemetry dataset, which captures realistic data from heterogeneous IoT networks, and a custom dataset. Six ML algorithms were tested, DT, RF, NB, LR, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). KNN achieved the best results with an accuracy of 99.98%, a precision of 99.95%, a recall of 99.97%, and F1-score of 99.96%.

Saheed and Arowolo [114] propose a cyber attack detection system for the IoMT environment based on a Deep Recurrent Neural Network (DRNN) and several supervised ML algorithms (RF, DT, KNN, Ridge Classifier (RC)). The system workflow starts with the preprocessing and normalization of the network data, then applies feature optimization using the bio-inspired Particle Swarm Optimization (PSO) algorithm to reduce dimensionality and select the most influential features. The evaluation is performed on the NSL-KDD dataset, covering several types of attack (DoS, probing, u2R). The PSO-RF model achieved the best performance with a maximum accuracy of 99.76%. In addition to high accuracy, the system reduces training time, which is essential for efficient deployment in IoMT environments. The authors plan to integrate blockchain technology to improve the security of future versions of the system.

Idrissi et al. [115] propose an IDS for IoMT devices based on Gradient Boosting Machine algorithms, including XGBoost, LightGBM, and CatBoost. Motivated by the COVID-19 pandemic and the growing dependence on IoMT devices for remote patient monitoring and home healthcare, the authors emphasize the urgent need to secure these devices, which are often vulnerable to cyberattacks. The proposed IDS was evaluated using multiple metrics, achieving detection rates greater than 99% in accuracy, recall, precision, and F1 score. Among the models tested, LightGBM demonstrated superior performance in all evaluated metrics, indicating its potential as an effective and fast solution for real-time IoMT intrusion detection. The study highlights the critical role of IoMT devices in managing healthcare surges and underscores the need for robust security measures to protect them from attacks.

Muhammad Naveed et al. [116] propose a hybrid IDS model for medical IoT devices, integrating normalization, feature selection (Pearson correlation) and SVM classification. Their approach achieves 99.3% accuracy on a standard dataset, outperforming existing methods. The model improves security while reducing processing costs. The authors recommend the future use of specific data and the exploration of advanced ML and DL techniques to optimize intrusion detection in IoMT.

Ghourabi [58] proposes a hybrid security model to protect healthcare systems against cyberattacks targeting IoT, IoMT, and Windows environments. The solution combines an optimized LightGBM-based intrusion detection system with a BERT-based Transformer malware detection model. Both models are trained on four diverse datasets to improve detection accuracy across different types of attacks. The approach achieves a high ROC-AUC score above 99%, demonstrating strong performance. This study highlights the effectiveness of combining ML techniques for comprehensive healthcare cybersecurity. Future work includes expanding malware coverage and developing integrated analysis methods to improve decision making. This research contributes valuable information on the security of complex medical device networks using AI-based methods.

Binbusayyis et al. [117] conduct a comparative evaluation of five ML algorithms (NB, KNN, DT, SVM, and ANN) for intrusion detection in IoMT networks. Using the public Bot-IoT dataset and Z-score normalization, the study shows that the DT achieves optimal performance with 100% accuracy, detection rate, false positive rate, and F1-score in binary classification. These results confirm the superiority of the DT for this task. This research fills a gap in the comparative evaluation of ML models specific to IoMT security and guides the selection of effective IDS systems. The authors plan to explore DL using algorithms to enhance the detection of more sophisticated attacks in the future.

Gupta et al. [118] propose a network intrusion detection model tailored for IoMT environments, using a tree-based classifier. The system combines an RF algorithm with robust feature scaling and dimension reduction techniques to efficiently process large-scale and heterogeneous medical data. The model achieves an average accuracy of 94.23% and an F1-score of 93.8%, while significantly reducing the classification time. Compared to LR, DT, and extra tree classifiers, it demonstrates superior efficiency and effectiveness. The methodology includes feature selection, encoding, and the use of SMOTE for data augmentation. This approach ensures scalability and real-time performance in e-healthcare systems. Future work involves integrating more attack types, experimenting with lightweight DL models, and refining dimensionality reduction methods to improve stability and AUC scores.

Zubair et al. [119] proposed a decentralized intrusion detection architecture based on DL to secure Bluetooth communications in smart healthcare systems, introducing BlueTack, the first public dataset dedicated to Bluetooth Classic and BLE attacks. Their IDS, evaluated using various algorithms (LR, DT, SVM, RF, NB, IF, KM, LOF, DNN) and preprocessing techniques such as Z-score normalization and feature selection methods (RFE, Chi-2), achieved 99.15% accuracy and a 99.6% F1-score in binary classification RF model, and a 97.73% F1-score in multiclass classification DT model. These results demonstrate the system's high performance and robustness against DoS, DDoS, and Bluesmack attacks targeting Bluetooth-enabled medical devices.

Vu-Duc Ngo et al. [120] conduct a comparative analysis of two key dimensionality reduction techniques-feature selection and feature extraction in the context of ML-based intrusion detection for IoT networks. Using the modern UNSW-NB15 dataset, they evaluated binary and multiclass classification scenarios. Their results show that feature selection generally outperforms feature extraction in terms of accuracy, precision, recall, and runtime, especially when the number of features $K \geq 8$. For instance, with DT and feature selection at $K = 8$ in binary classification, they achieved 87.07% precision, 87.87% recall, 87.07% accuracy, and F1-score of 87.47%. However, for very small feature sets $K \leq 4$, feature extraction is more robust and accurate. DT performs best with feature selection, while MLP (Multi-Layer Perceptron) excels with feature extraction. The study provides practical guidelines for selecting the appropriate feature reduction method based on detection goals and data constraints. Future work will investigate these findings across other datasets and explore DL-based classifiers.

Norouzi et al. [121] proposed a novel IoMT framework to secure patient medical data against cyberattacks, combining a hybrid genetic algorithm-based Random Forest (GA-RF) model for intrusion detection. Their approach demonstrated outstanding performance, achieving up to 99.99% accuracy and 100% precision and recall on the NSL-KDD and UNSW-2018-IoT-Botnet datasets, outperforming other ML algorithms.

Dadkhah et al. [33] applied an IDS using ML models on a newly developed dataset designed for binary, multiclass, and multi-type intrusion detection problems. Their results demonstrated high performance across the evaluation metrics, and their work provides the research community with a broader perspective for developing new techniques thanks to this large and comprehensive dataset.

Hernandez-Jaimes et al. [122] proposed an approach that combines Nilsimsa fingerprinting with ML models to detect ransomware-spreading behavior in IoMT network traffic. Their method addresses the feature dependency issue of traditional ML-based IDS by efficiently representing network flows and was evaluated using the ICE dataset which contains both clean traffic and ransomware attack scenarios and the CICIoMT2024 dataset, achieving high precision and F1-scores.

Areia et al. [123] developed a new dataset named IoMT-TrafficData, which is divided into three main parts: a flow dataset, a packet dataset, and a BLE dataset. However, their work focused on implementing an IDS only for the flow and packet parts, using ML algorithms to detect attacks in both binary and multiclass scenarios. Their models achieved high evaluation metrics, although a slightly high false positive rate negatively affected the overall prediction performance. As future work, they suggest applying ML or DL techniques to the BLE part of the dataset and exploring ways to minimize false negatives.

Alalwany et al. [124] proposed a novel IDS implemented within a Kappa Architecture framework, using a stacking ensemble learning approach that outperforms single algorithms such as AdaBoost, NB, and LR. This stacking technique, which belongs to ensemble learning methods, allows multiple base classifiers to be trained in parallel on the same dataset, and then combines their outputs (e.g., by averaging) to achieve higher accuracy while leveraging the strengths of all the models. Although this method increases training time, it remains acceptable for real-time IoMT applications thanks to the efficiency of the Kappa Architecture.

Lipsa et al. [125] address critical IoMT security challenges by proposing an interpretable feature selection technique combined with an explainable AI-based IDS. The main goal is to enhance attack detection accuracy while ensuring decision transparency, which is crucial for security analysts and healthcare practitioners. The authors employed Recursive Feature Elimination (RFE), Mutual Information (MI), and Lasso Feature Selection (LFS) to identify the most relevant features, reduce dataset dimensionality, and retain high detection performance. By leveraging SHAP values, they provide clear explanations of feature importance, overcoming the black-box nature of traditional IDS models. Their approach achieved 99% accuracy with RF, outperforming XGBoost, SVM, and DT, while improving efficiency and interpretability. However, they note that computational complexity may limit real-time use, and further validation in large-scale IoMT environments is needed. Future work will focus on real-time IDS integration, edge computing optimization, hybrid AI models, and extending the approach to other IoT domains like smart cities and industrial IoT. Overall, the proposed method strengthens IoMT security and detection accuracy while ensuring greater model explainability, making it practical for real-world deployment.

Sheik Abdullah1 et al. [126] proposed a dataset generation method using an open-source tool called IoT-Flock. This approach produced the Healthcare-IoT dataset, which includes two classes: normal and attack. The authors applied several ML methods to this dataset, achieving the best result with XGBoost (XGB), which reached an accuracy of 99.67%. This work demonstrates the effectiveness of the IoT-Flock tool for generating realistic datasets to train and evaluate ML modules, highlighting its practical value whenever reliable IoT security datasets are needed.

Yacoubi et al. [127] applied ML modules for attack detection and developed an explainable IDS capable of clarifying how the model makes its decisions. They also proposed a novel approach that reduces the number of features by applying SHAP to select the top 15 features. This technique succeeded in maintaining high accuracy while significantly reducing training time across all the classification models used. Among them, RF achieved the best performance, reaching an accuracy of 99.87%.

Nasayreh et al. [128] propose an intelligent framework for anomaly and cyberattack detection in the IoMT. Their approach is based on a hybrid LSTM-PCA-KNN model, where LSTM is used for temporal feature extraction, PCA for feature reduction and

Table 6

DL model classifications for IoMT applications.

Year	Ref	DL Algorithm	Normalization	Feature Selection	Best Clf	Class Type	Accuracy %	Precision %	Recall %	F1-score %	Dataset
2020	[129]	KNN, NV, RF, SVM, DNN	Standard-scalar	–	DNN	Binary	99.9	–	93.2	–	NSL-KDD
2020	[130]	RF, KNN, SVM, ANN	–	–	ANN	Binary	93.42	–	–	–	WUSTL EHMS
2023	[131]	CNN, LSTM	–	–	CNN-LSTM	Binary	99	99	98	98	WUSTL EHMS
2023	[132]	CNN, LSTM	Z-normalization	MI	CNN-LSTM	Multiclass	97.63	98.47	97	97.73	CIC-IDS2017
2023	[7]	QDNN	–	–	QDNN	Binary	100	100	100	100	WUSTLEHMS-2020 ICU
		MLP, LSTM, CNN, CNN-LSTM	–	–	MLP	Binary Multiclass	99.99 96	– –	– –	– –	UNSW-NB15, Edge IIoTset
2024	[134]	DAN2 NB, SVM, LR, DT, DNN	Standard-scalar	BDA	DAN2	Binary	98.92	98.50	98.68	97.90	TON-IoT
2024	[135]		–	–	DNN	Binary (MQTT) Binary (COAP) Multiclass (MQTT) Multiclass (COAP)	98.9 99.3 99.9 99.9	99.0 99.2 99.3 99.9	98.9 99.5 99.4 100	99 99.4 99.3 99	Generated by, IoT-Flock [136]
2025	[137]	RF, SVM, LR, VAE, XGBoost, LightGBM, FNN, TNN	–	–	VAE	Binary	91.61	91.69	91.61	91.60	WUSTLEHMS-2020
2025	[138]	CNN-LSTM, CNN, LSTM	–	Pearson correlation	CNN-LSTM	Multiclass	99.87	98.5	98.64	98.54	CIC-DDoS2019

optimization, and KNN for traffic classification. The model is evaluated on four real-world healthcare datasets (TON-IoT, ECU-IoHT, ICU, and WUSTL-EHMS). Experimental results demonstrate outstanding performance, achieving 99.9% accuracy, recall, precision, and F1-score on the WUSTL-EHMS dataset, and consistently outperforming conventional models such as LSTM, KNN, SVM, and Naïve Bayes. The proposed framework effectively enhances the security of IoMT devices in hospital environments.

5.3. IDS based on DL models in IoMT

This section explores IDS in the IoMT that leverage DL models, highlighting their architectures, techniques, and effectiveness in identifying complex cyber threats. Table 6 summarizes the collection of studies conducted in this area.

Swarna Priya et al. [129] compared several ML and DL algorithms (KNN, NB, RF, SVM, and DNN) for intrusion detection. Using Standard-Scaler normalization and no specific feature selection, the DNN achieved the best results for binary classification with 99.9% and 93.2% recall on the NSL-KDD dataset.

HADY et al. [130] developed an IDS for healthcare by using both network traffic data and patient's biometric information at the same time to detect attacks more effectively. They tested their method on an Enhanced Healthcare Monitoring System (EHMS) with several body sensors, creating a real dataset of over 16,000 records containing both normal traffic and MITM attacks. They evaluated four ML models (RF, KNN, SVM, ANN) with three different sets of features. The results showed that using both types of data improved the AUC by up to 25%. The ANN gave the best results, and prediction time was not significantly affected. Still, the authors mentioned that the system could be improved further.

Ravi et al. [131] introduced a DL-based intrusion detection approach specifically designed for IoMT systems. This technique leverages both network traffic characteristics and patient biometric signals, allowing the model to develop rich internal representations through its hidden layers. To address the issue of imbalanced class distribution, a cost-sensitive training method was employed. Additionally, the model incorporates Convolutional Neural Network (CNNs) and Long Short-Term Memory (LSTMs) to extract spatial and temporal patterns from the input data. The effectiveness of this approach was demonstrated using three well-known datasets WUSTL EHMS 2020, SDN-IoT, and KDDCup-99 achieving 95% accuracy with network data alone, 89% with biometric features alone, and up to 99% when both data sources were combined.

The paper by Faruqui et al. [132] introduced SafetyMed, a novel IDS for IoMT, which combines CNN and LSTM to detect intrusions from both image-based and sequential network traffic. This approach provides robust protection against twelve different types of attacks and achieves an effective balance between detection rate and false positive rate. To enhance model performance, the authors employed MI for feature selection. SafetyMed demonstrated high reliability with an average accuracy of 97.63%, a precision of 98.47%, and a low false positive rate of just 0.71%, highlighting its potential for securing resource-constrained medical environments.

Al-Hawawreh and Hossain [7] proposed a privacy-aware framework for cyberattack detection in IoMT systems, addressing key challenges related to secure data collection, privacy preservation, and heterogeneity of medical data. Their approach integrates distributed cloud storage with privacy-preserving mechanisms, emphasizing the importance of data fusion (DF) across heterogeneous IoMT sources. To mitigate privacy risks during data fusion, the framework applies T-closeness, ensuring strong anonymization by maintaining distributional similarity of sensitive attributes. A contractive deep autoencoder with differential privacy is then used to

reduce data dimensionality, followed by a quantum deep neural network (QDNN) for efficient cyberattack detection. Evaluated on real-world IoMT datasets, the system achieved over 99% detection accuracy while preserving sensitive data, outperforming conventional detection models. The study highlights the synergy between quantum learning and privacy-preserving techniques for securing IoMT ecosystems, and suggests further exploration of adaptive quantum models and broader data fusion strategies under varying privacy budgets.

Sy et al. [133] proposed a DL-based IDS to address the growing threats targeting connected medical devices, particularly DDoS and botnet attacks. Their approach leverages four DL architectures MLP, LSTM, CNN, and a hybrid CNN-LSTM to detect anomalies in IoT network traffic. Evaluated on the UNSW-NB15 and Edge-IIoTset datasets, their system achieved 99.8% accuracy in binary classification and 96% in multiclass classification, with a false alarm rate below 2%, demonstrating the effectiveness of DL in enhancing the cybersecurity of medical IoT systems. The study further highlights the potential of integrating edge computing, FL, steganography, and blockchain to improve adaptability, privacy, and resilience against adversarial threats.

Raid Mohsen Alhazmi [134] proposed an IDS for the IoMT based on the BDA-DAN2 model. The methodology includes several stages: dataset collection, data preprocessing using standard scalar techniques, feature selection via the DL-based binary dragonfly algorithm (BDA) algorithm, and classification using the Artificial Neural Networks2 (DAN2) algorithm to enhance system accuracy. DAN2 is an advanced neural network model that builds upon traditional ANN by introducing dynamic learning capabilities and enhanced feature discrimination. It is particularly effective in complex classification tasks, such as intrusion detection, where high precision and adaptability are required. The IoMT dataset was used to train and test the model. Performance evaluation was conducted using accuracy, precision, recall, and F1-score. The proposed system achieved impressive results: 98.92% accuracy, 98.50% precision, 98.68% recall, and a 97.90% F1-score. These metrics outperformed various existing models in the literature, such as KNN, SVM, Safety-Med, PSO-DNN, FST-LSTM, XMedNN, PSO-Adaboost, DBN-botnet, and PCA-MLP. The study demonstrates that the BDA-DAN2-based IDS is highly effective for IoMT environments, where secure communication between smart medical devices is critical. Future work may enhance model performance by enlarging the dataset and incorporating multi-class classification techniques.

Lerina Aversano et al. [135] proposed an explainable anomaly detection framework for IoT medical networks using synthetic traffic generated by the IoT-Flock tool. The methodology simulates IoT traffic in a smart healthcare environment, supporting both CoAP and MQTT protocols, and incorporates normal as well as malicious packets representing four types of attacks. The process involves dataset generation through IoT-Flock, feature extraction at the application layer, and classification using traditional ML algorithms, DL models with multiple hidden layers, and XAI techniques to derive human-interpretable decision rules. The binary classification experiments achieved nearly 100% accuracy for CoAP traffic, while multinomial classification also yielded high performance, with CoAP packets classified more accurately than MQTT packets. The study demonstrates that IoT-Flock-generated synthetic data can effectively train and evaluate models for anomaly detection in medical IoT scenarios. Future work aims to validate the approach by training on synthetic traffic and testing on real-world labeled medical IoT datasets, incorporating per-flow analysis, and applying feature selection to maintain accuracy while reducing computational overhead.

Mamta Kumari et al. [137] proposed a DL-based intrusion detection framework for the IoMT targeting MiTM attacks in smart healthcare environments. The methodology leverages real-time data from the St. Louis Enhanced Healthcare Monitoring System (WUSTL-EHMS) and applies a diverse set of ML and DL algorithms, including Variational Autoencoders (VAEs), RF, SVM, Gradient Boosting Machines (XGBoost and LightGBM), LR, Feedforward Neural Networks (FNN), and MLPClassifier. The models are evaluated using accuracy, precision, recall, and F1-score. VAEs achieved the best performance, with 91.61% accuracy, effectively capturing complex threat patterns, while RF and SVM also demonstrated strong detection capabilities. LightGBM yielded high precision but slightly lower recall and F1-score, and LR maintained balanced precision and recall despite lower accuracy. FNN and MLPClassifier proved to be reliable alternatives, and XGBoost matched VAEs in overall classification performance. The study highlights the critical role of AI-driven IDS in securing IoMT environments, emphasizing adaptability to evolving cyber threats. Future research directions include developing lightweight, resource-efficient models suitable for deployment on IoMT devices, ensuring both computational feasibility and high detection accuracy to safeguard patient safety and data privacy.

BERGUIGA et al. [138] proposed HIDS-RPL, a hybrid IDS combining CNN and LSTM to detect DDoS attacks in IoMT networks using the Routing Protocol for Low-Power and Lossy Networks (RPL). RPL is widely used in 6LoWPAN-based IoT applications to manage dynamic topologies with constrained devices, but it is vulnerable to DDoS attacks exploiting control messages like DIO, DIS, and DAO. Trained and tested on the CIC-DDoS2019 dataset, the proposed model achieved 99.87% accuracy, 98.5% precision, 98.64% recall, and an F1-score of 98.54%.

5.4. Deploying AI models for real-time threat detection

Model deployment is a crucial phase for evaluating any proposed architecture within the context of IoT. This deployment involves several critical constraints, primarily related to the hardware used, which must be capable of supporting real-time applications. Among these constraints are limited memory, which restricts the size of executable models, and low computational power of embedded devices. In addition, energy consumption is a significant concern, as each model consumes power differently depending on its structure and the complexity of its decision-making process [139]. The response time of the model, that is, its ability to make decisions quickly, is also a key factor. Deploying the model directly on the IoT device enhances inference precision and enables real-time responsiveness by eliminating the need for multiple intermediate processes that would otherwise increase latency in decision-making. All of these constraints directly impact the selection of the appropriate hardware platform (microcontroller, processor, etc.) as well as the choice of deployment tools. For instance, TinyML is a widely used framework for deploying ML models on resource-constrained

Table 7

Comparative analysis of FL-Based IDS in IoMT.

Year	Ref	FL Partitioning	Learning Algorithm	Aggregation Strategy	FL Framework	Accuracy %	Privacy	Dataset
2021	[146]	Horizontal	DNN	FedAvg	TensorFlow (Keras) PyTorch	[88.27–95.14] 91.40	LOW	CICIDS2017 ECU-IoHT
2023	[147]	Horizontal	DNN	FedAvg	–	98.47	Medium	wustl-ehms-2020
2024	[148]	Horizontal	EFL-LSTM	FedAvg	MATLAB 2024a	97.16	LOW	ECU-IoHT
						99.69 99.68 99.89		EHMS-2020 M-En M-IoT-Env
2024	[149]	Horizontal	Meta-learning Reinforcement learning	FedAvg	Flower	99.40	LOW	ToNIoT
2025	[151]	Horizontal	CNN	FedAvg	Flower	95.78	LOW	CICIoMT2024
2025	[152]	Horizontal		differential privacy	PyTorch	Medium	–	–

embedded devices. In Abadade et al. [140], TinyML has been successfully applied in the medical field to enable real-time detection of respiratory diseases using lung sound recordings on low-power digital stethoscopes. This study demonstrates the feasibility of deploying lightweight ML models directly on medical devices, ensuring both efficiency and responsiveness without reliance on cloud infrastructure. It highlights the growing potential of TinyML for developing accessible and portable diagnostic solutions, particularly in remote or underserved areas. In Hamdouchi and Idri [141], Fusco et al. [142], Fatemeh et al. [143], researchers developed IDS systems based on TinyML, specifically designed for resource constrained environments such as IoT devices, to detect cyberattacks. These lightweight intrusion detection models were optimized to operate on microcontrollers with limited memory and processing power, enabling real-time threat detection at the network edge without relying on cloud-based infrastructure. The researchers evaluated the feasibility and performance of their models by deploying them on real-world datasets and reported promising evaluation metrics such as accuracy, precision, recall, and latency. Their work highlights the potential of TinyML to bring intelligent and autonomous threat detection capabilities directly to embedded systems, contributing to decentralized and scalable cybersecurity solutions for the IoT ecosystem. In Soe et al. [144], Soe et al. presented the impact of the feature selection phase during preprocessing, highlighting its influence on memory usage as well as training and testing execution time. They implemented a lightweight intrusion detection solution deployed on a Raspberry Pi for attack detection. Other deployment options also exist, such as TensorFlow Lite, which is designed to run DL models on mobile and embedded devices. It optimizes model size and latency, making it suitable for real-time inference while preserving acceptable accuracy and low resource consumption. In Oishee et al. [145], Oishee et al. proposed a lightweight DL solution for detecting respiratory diseases using the ICBHI dataset. The CNN-LSTM model achieved the best results with 96% accuracy and F1-score. The model was optimized with pruning and quantization for deployment on resource-constrained devices and integrated into the RespiScan smartphone app for real-time respiratory health monitoring. This approach highlights AI's potential for early disease detection and efficient health management in limited-resource environments (Table 7).

5.5. Federated learning (FL)

FL is a decentralized ML approach that enables multiple devices or organizations to collaboratively train an AI model without directly sharing their sensitive data [153]. Each participant trains the model locally on its own data and sends only the updated parameters (e.g., model weights) to a central server, which aggregates them [154]. This enhances data privacy and reduces risks associated with transferring sensitive information, making it highly suitable for sectors like healthcare and the IoMT [155]. Fig. 17 illustrates a typical FL architecture, where multiple hospitals (e.g., Hospital 1 and Hospital 2) train local models on their respective datasets. Each participant computes model updates (Local Model) and sends only the parameters (e.g., weights) to a central server (Server). The server aggregates these contributions to refine the Global Model, which is then redistributed to all participants. This iterative process enables collaborative model improvement without direct data sharing. In Singh et al. [156], the authors address the security challenges of smart hospitals and the IoMT ecosystem, which remain vulnerable to cyber threats despite relying on large-scale AI and cloud infrastructures. To enhance data privacy and intrusion detection, they propose a Dew-Cloud architecture that implements hierarchical federated learning (HFL) combined with a hierarchical LSTM (HLSTM) model. This HFL-HLSTM approach effectively detects attack vectors in IoMT traffic, outperforming existing state-of-the-art intrusion detection methods. The model's performance, validated on TON-IoT and KDD datasets, demonstrates high accuracy (99.31%), precision (98.97%), recall (98.24%), and f1-score (98.58%), with promising results for both binary and multi-class classification tasks. The study concludes that this framework improves security, reliability, and user trust in IoMT systems, while suggesting future enhancements through Gurobi optimization to further boost accuracy and reduce latency. In Ioannou et al. [157], the authors propose an energy-efficient IDS to secure IoMT networks through a three-stage approach: classification of known attacks using an Enhanced RF (achieving 99.98% accuracy), anomaly detection with a One-Class SVM (99.7% accuracy), and continuous updates via FL to preserve data privacy. Validated on an IoT system with a Raspberry Pi IoMT gateway, the solution demonstrates reliable detection of both known and unknown attacks while optimizing resource usage. Future work aims to enhance the scalability and efficiency of this approach within connected healthcare infrastructures.

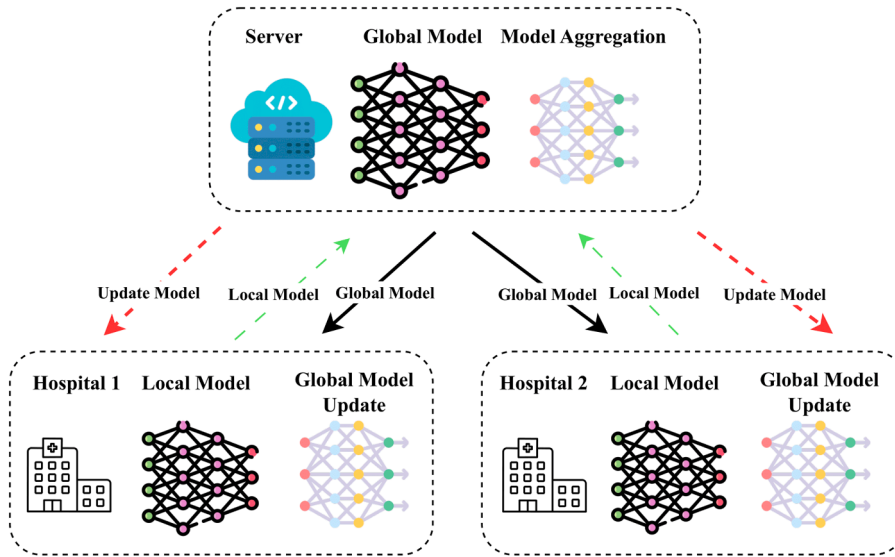


Fig. 17. FL for IoMT: Global Model Aggregation from Distributed Hospital Data.

In the context of intrusion detection, FL enables the deployment of collaborative IDS across distributed IoMT and IoT infrastructures while preserving data locality and data privacy. FL is not a single paradigm but encompasses several learning types, mainly *horizontal FL*, *vertical FL*, and *combined (or hybrid) FL*. Horizontal FL is applied when participants share the same feature space but hold different data samples, which is common in IoMT gateways monitoring similar network traffic features. Vertical FL is used when participants share the same samples but have different feature sets, such as combining network-level and device-level attributes. Combined FL addresses more complex scenarios where both the feature space and data samples differ across participants [158]. Unlike centralized IDS, FL-based IDS can continuously learn from diverse attack patterns observed at different sites without exposing sensitive network traffic. This paradigm is particularly effective for handling emerging attacks, concept drift, and heterogeneous data distributions across medical environments. Table 11 summarizes the main research works that employ FL for IDS in IoMT and IoT environments for cyber-attack detection. However, FL-driven IDS also face challenges such as non-IID data, communication overhead, straggler clients, model poisoning, and aggregation robustness, which must be addressed to ensure reliable and secure intrusion detection in IoMT systems [159].

5.6. IoT Datasets

In recent years, the proliferation of connected devices has generated a wide range of datasets dedicated to the IoT. These datasets are essential resources for researchers and practitioners aiming to design and assess effective intrusion detection solutions. Table 8 presents a selection of IoT-related datasets, emphasizing their importance for developing and evaluating intrusion detection models that can also be adapted to strengthen the security of IoMT environments.

5.7. IoMT attack datasets

The effectiveness of IDS in the IoMT largely depends on the availability and quality of relevant datasets used for training and evaluation. IoMT environments generate highly sensitive and heterogeneous data, which makes it essential to use datasets that reflect real-world traffic patterns, device behavior, and various types of cyberattacks specific to healthcare systems. Several datasets have been proposed to simulate or capture network traffic within IoMT infrastructures, incorporating both benign and malicious activities. These datasets are crucial for developing AI-based IDS models, enabling them to accurately detect intrusions such as spoofing, DoS, MITM attacks, and data exfiltration attempts. Additionally, they support performance benchmarking and the comparative evaluation of detection techniques across different platforms. This section provides an overview of commonly used IoMT-related attack datasets, highlighting their characteristics, attack coverage, and suitability for AI-driven intrusion detection research.

Table 9 presents a comparative overview of widely used IoMT-related attack datasets. For each dataset, key characteristics are highlighted, such as the year of release and the classification type (binary or multiclass), the range of cyberattacks covered, and the ML models typically applied. We also detail the underlying technological context in which the dataset was generated (e.g., simulated or real IoMT environments), the network protocols involved, the dataset size in terms of the number of rows, and the number of features available for model training. This comprehensive comparison helps identify the strengths and limitations of each dataset, enabling researchers to select the most appropriate one for developing and evaluating AI-driven IDS in healthcare-focused IoMT scenarios. By aligning dataset properties with specific research objectives, practitioners can enhance detection accuracy, reduce false positives, and improve the generalizability of their IDS models.

Table 8

Comparative overview of IoT attack datasets.

Year	Dataset	Binary/ Multiclass	Attack	Tools	Size of rows	Number of features
1999	KDD CUP [160]	Both	DoS, R2L, U2R, probing	–	4,898,430	41
2009	NSL-KDD [161]	Both	–	–	125,973	41
2015	UNSW-NB15 [162]	Binary	Fuzzers, Shellcode, Worms, Generic, Reconnaissance, Exploit, Backdoor, Analysis	Argus, Bro-IDS Hydra, Medusa, Ncrack, Nmap, Metasploit, LOIC, HOIC, Hulk, GoldenEye, Slowloris, Nessus, Slowhttptest, Grum, Windigo, High Orbit, Ion Canon, Low Orbit Ion Canon, DDoSIM, Nikto, WebScarab	2,540,044	49
2017	CIC-IDS2017 [163]	Both	Brute Force, DoS Hulk, Infiltration, Port Scan, DDoS, LOIT, XSS, SQL Injection, DoS GoldenEye, DoS, Slowhttptest, DoS Slowloris, SSH-Patator, FTP-Patator	–	2,830,540	80
2018	CSE-CIC-IDS2018 [164]	Both	Bot, Brute Force Web, Brute Force-XSS, DDoS HOIC, LOIC-UDP, LOIC-HTTP, FTP-BruteForce, Slowloris, SlowHTTPTest, Hulk, GoldenEye	–	8,392,401	75
2018	Bot-IoT [165]	Both	DoS, DDoS, OS and Service Scan, Theft	Argus, Tshark, Nmap, Ostinato, Hping3, Node-RED, Metasploit, GoldenEye, Xprobe2	73,912,589	42
2020	MQTT-IoT-IDS [166]	Both	Aggressive scan, UDP scan, Sparta SSH brute-force, MQTT brute-force attack	VLC, Nmap, MQTTPWN	225,711	68
2020	TON-IoT [167]	Both	Backdoor, DoS, DDoS, Injection, MITM, Scanning, Ransomware, XSS, Password	Nmap	461,043	44
2022	RT-IoT2022 [168]	Both	ARP poisoning, DoS SYN, Hping, brute-force SSH, Slowloris distributed, DDoS, Nmap	–	123,117	83
2022	CIC-IoT2022 [169]	Both	Flood DoS, RTSP brute-force	Hydra, Nmap	30,000	80
2023	CIC-IoT2023 [170]	Both	ACK fragmentation, UDP flood, Slowloris, ICMP flood, RSTFIN flood, PSHACK flood, HTTP flood, UDP fragmentation, TCP flood, UDP Plain, SynonymousIP flood, ARP spoofing, DNS spoofing, SYN flood, OS scan, UDP flood, Ping sweep, Vulnerability scan, Port scan, SQL injection, Command injection, Backdoor malware, Uploading attack, XSS, Browser hijacking, GREIP flood	Hping3, UDP flood, Slowloris, Nmap, fping, Vulscan, Ettercap, DVWA, Wireshark	46,686,579	44

Table 9
Comparative overview of IoMT attack datasets.

Year	Dataset	Binary multiclass classification	Attack	ML models	Technology	Protocols	Size of row	Number of feature
2019	ICE [171]	Both	Wannacry, Petya, BadRabbit and, PowerGhost	OC-SVM, LOF, IF, NB, RF, NN	WiFi	–	10 000 000	520
2020	WUSTL EHMS [130]	Both	Spoofing attacks and Data, alteration	RF, KNN, SVM, ANN	WiFi	–	16 318	42
2021	ICU [18]	Binary	MQTT Publish Flood, MQTT Authentication, Bypass, MQTT Packet, Crafting Attack, COAP, Replay Attack	NB, KNN, Adaboost, LogR, DT KNN, LOF, COF, aLOCI, RPCA,HBOS , INFLO, CBLOF,	WiFi	MQTT COAP	188 697	–
2021	ECU-IoHT [71]	Both	DoS, ARP spoofing, Smurf, Nmap	CMGOS,LIBSVM	WiFi	–	111 207	9
2022	BlueTack [119]	Both	Bluesmack, DoS, DDoS, MiTM MiTM, DDoS, Ransomware,	–	BR/EDR, and BLE	–	30 628	22
2024	WUSL HDRL [172]	Both	Buffer Overflow ARP spoofing, Ping, Sweep, Recon VulScan, OS Scan, Port Scan, MQTT Malformed Data, MQTT DoS Connect flood, MQTT DoS Publish flood, MQTT DDoS Connect flood, MQTT DDoS Publish flood, DoS TCP, DoS ICMP, DoS SYN , DoS UDP, DDoS TCP, DDoS ICMP, DDoS SYN, DDoS UDP	–	5G	–	145 123	77
2024	CICIoMT2024 [33]	Both	Recon, Bluetooth Injection, Bluetooth Recon, Apache killer, R-U-Dead-Yet, ARP Spoofing	LR, AdaBoost, RF, DNN DT, RF, NB, LogR, SVM, DNN	WiFi, Bluetooth	MQTT	8 775 013	44
2024	IoMT-TrafficData [123]	Both			WiFi, Bluetooth	MQTT	974 630	–

Table 10
Overview of explainable intrusion detection studies in the IoMT domain.

Year	Ref	AI Approach	Explainable Module	Attack	XAI Approach	Dataset
2022	[182]	XGBoost, auto-encoder	XGBoost	Probe, Attack	SHAP	NSL-KDD
2023	[183]	RF, DT, SVC, XGB RF, ADA, DNN, SVM, KNN, MLP,	Voting	malicious	SHAP	WUSTL-EHMS2020 RoEduNetSIMARGL2021, CICIDS-2017, NSL-KDD
2024	[184]	LightGBM DT, RF, KNN, SVM, ANN, MLP,	All	malicious	SHAP, LIME	WUSTL-EHMS2020
2025	[177]	XGBoost	XGBoost, RF	malicious spoofing	LIME	CICIoMT2024
2025	[185]	TNN	TNN	DoS,MQTT, DDoS,Recon, Spoofing	SHAP, LIME	CICIoMT2024 NSL-KDD, UNSW-NB15, ToN-IoT, WUSTL-EHMS
2025	[187]	ANN	ANN	malicious	SHAP	

Table 11
Comparative analysis of FL-Based IDS in IoMT.

Ref	Year	FL Partitioning	Learning Algorithm	XAI Method	Visualization Type	Explanation Scope	Dataset
[188]	2025	Horizontal	CNN-Bi-LSTM	SHAP	Pixel-level	Local	CL2-IDS
[152]	2025	Horizontal	DNN CNN	SHAP	Beeswarm Plot Summary Plot Waterfall Plot Decision Plot	Global Global Local Local	wustl-ehms-2020 ECU-IoHT CIC-IDS2017 N-BalIoT EdgeIoTset
[189]	2025	Horizontal	LSTM-AE	SHAP			

6. Explainable AI for understanding cyberattacks in medical IoMT environments

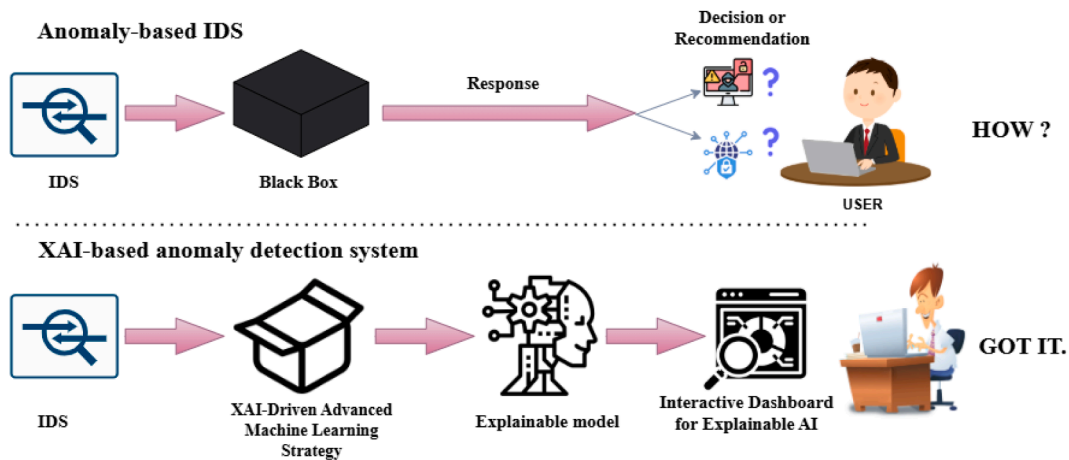


Fig. 18. From black box to transparent AI: XAI-Driven Anomaly Detection Workflow.

A major challenge in the adoption of AI, particularly in the field of cybersecurity, is the lack of transparency in decision-making. Traditional AI models act as “black boxes” that generate results without clear explanations, which can reduce trust among security experts [173]. Researchers in [174] presented the four possible types of explainability: 1) data explainability, 2) model explainability, 3) post-hoc explainability, and 4) assessment of explanations. In Belle and Papantonis [175], a comprehensive map of explainability approaches is presented, detailing the various categories of explainability, the types of models associated with each category, and the commonly used techniques. The application of XAI in the medical field has grown significantly, particularly in supporting disease interpretation [176], and it has also contributed to the advancement in explaining attacks on medical machines [177]. The Fig. 18 illustrates a clear comparison between traditional black-box anomaly detection systems and explainable AI-based anomaly detection systems. In the traditional approach, the IDS processes data and feeds it into a black-box model, which generates a response or decision. However, this process lacks transparency, leaving the user uncertain about how or why certain decisions or recommendations were made, as the internal workings of the black-box are opaque. This leads to challenges in trust, interpretability, and actionable insights. In contrast, the XAI-based anomaly detection system incorporates an advanced ML strategy driven by explainability. The output is an explainable model that feeds into an interactive dashboard, providing the user with clear, interpretable insights and detailed reasoning behind the detection results. This not only helps users understand the decisions better but also facilitates informed decision-making and confidence in the system’s outputs, making it a significant advancement over the traditional black-box approach. Several studies conducted in this direction are summarized in Table 10, highlighting diverse explainable techniques integrated into IDS frameworks, their use cases, benefits, and the improvements achieved in detection transparency and system accountability.

6.1. Four principles of explainable AI

To foster trust, transparency, and accountability in AI systems, the National Institute of Standards and Technology (NIST) has outlined four fundamental principles of XAI [178]. These principles serve as essential guidelines for the development and evaluation of AI models that aim to provide understandable, reliable, and ethical outcomes. By adhering to these principles, AI systems particularly in sensitive domains such as cybersecurity, healthcare, and finance can offer users greater insight into their decision-making processes, ultimately improving user trust and system adoption. The four NIST principles are as follows:

- **Explanation:** The system should be able to deliver evidence or reasoning for all outputs.
- **Meaningful:** The explanation must be understandable to the intended user.
- **Accuracy:** The explanation should reflect the actual process of the system and provide a faithful representation of its decision-making.
- **Knowledge Limits:** The system should identify cases where it is likely to fail or when it is operating outside its intended scope.

6.2. Explaining decisions in IoMT IDS: Regulatory implications under GDPR and HIPAA

In the context of IoMT, where sensitive patient data is processed and critical security decisions are made, regulatory compliance plays a vital role in ensuring transparency and accountability. Both the GDPR in the European Union and the HIPAA in the United States impose obligations that support the integration of XAI into IDS. GDPR, particularly under [179], grants individuals the right not to be subject to decisions made solely by automated processing without meaningful explanation, thereby encouraging the use of interpretable ML models. Furthermore, of the GDPR establishes the principle of transparency [180], which requires that all

personal data processing be conducted in a lawful, fair, and transparent manner. This implies that if an algorithm makes a decision affecting an individual, that person must be able to understand how their data was used and why a specific outcome occurred. On the other hand, HIPAA [181], while not explicitly addressing AI, mandates strict data protection and grants patient's rights to access and understand their medical information, which indirectly supports the need for interpretability in automated clinical and security decisions. Consequently, adopting XAI-driven IDS in IoMT not only enhances user trust and interpretability but also aligns with legal frameworks that demand transparency, fairness, and informed decision-making in data-driven environments.

6.3. XAI-enabled federated IDS for IoMT

The increasing adoption of FL in IDS for the IoMT has significantly improved data privacy and collaborative intelligence across distributed healthcare environments. However, the black-box nature of most deep learning models deployed in FL-based IDS remains a major challenge in safety-critical medical applications, where transparency, trust, and regulatory compliance are essential. To address this limitation, recent research efforts have focused on integrating XAI mechanisms into federated IDS architectures. XAI-enabled federated IDS aim to provide interpretable and trustworthy security decisions while preserving the privacy advantages of FL. By embedding local and global explanation models (e.g., SHAP, LIME, Grad-CAM, and attention-based explanations) within federated architectures, researchers enable medical institutions to not only detect cyber attacks but also understand the underlying decision-making process of the IDS. This interpretability is crucial for clinical validation, incident forensics, model debugging, and compliance with stringent regulations such as GDPR and HIPAA. Moreover, the integration of XAI into federated IDS enhances collaboration among heterogeneous IoMT stakeholders by facilitating the sharing of explainable knowledge instead of opaque model parameters alone. This paradigm supports human-in-the-loop security monitoring, improves operators' confidence in automated detection, and contributes to the deployment of reliable, transparent, and ethically aligned cybersecurity solutions for next-generation IoMT infrastructures.

7. Next-generation AI-powered security operations center (SOC) for healthcare cybersecurity

Recent studies have highlighted the growing vulnerability of healthcare systems to sophisticated cyberattacks, driven by the high value of medical data and the critical nature of healthcare services. Traditional Security Operations Centers (SOCs), while essential for monitoring and incident response, are increasingly seen as insufficient in facing modern, fast-evolving threats. As a result, researchers and practitioners are shifting toward Next-Generation SOC, which integrate AI to enhance detection, correlation, and response capabilities [190,191]. A key component of this evolution is the use of Security Information and Event Management (SIEM) platforms, which provide centralized log collection, normalization, and real-time analysis of security data. Modern SIEM tools such as Elastic Stack (ELK) [192], Splunk [193], and IBM QRadar [194] are often extended with ML models and anomaly detection techniques to identify patterns that traditional rule-based systems may miss. For instance, ELK (Elasticsearch, Logstash, Kibana) is widely used in SOC environments for building custom dashboards, correlating multi-source logs, and integrating with intrusion detection systems like Suricata or Snort. In parallel, AI-powered intrusion detection systems such as Darktrace leverage unsupervised ML to model normal behavior across complex networks and autonomously detect subtle anomalies or emerging threats, making them a critical asset in proactive threat hunting and real-time defense [195]. An important foundational study examined the evolving role of SOC in managing the growing presence of IoT devices within network infrastructures [190]. The widespread deployment of IoT and ad hoc wireless connectivity introduces additional challenges for SOC, including the need for real-time visibility, accurate asset management, and understanding diverse IoT protocols. This study proposed a framework for integrated security monitoring and orchestration in hybrid IT-IoT environments. It highlighted classifications of IoT specific attacks and suggested phased approaches for integrating IoT into SOC operations, including temporary outsourcing of IoT monitoring, establishing specialized analyst teams, and full integration of IoT data within SIEM platforms. The work also emphasized the crucial role of automation and ML to enable scalable and effective security operations amid increasing infrastructure complexity. Another essential component in a SOC is Endpoint Detection and Response (EDR), a system dedicated to monitoring and protecting endpoints such as computers, servers, or gateways equipped with an operating system. EDR continuously collects data on the activities and behaviors of endpoints to quickly detect any suspicious or malicious activity [196]. Thanks to advanced algorithms, including artificial intelligence and behavioral analysis, it is capable of identifying sophisticated threats like ransomware, which attempt to encrypt or block critical data. In addition to detection, EDR also facilitates rapid incident response by enabling actions such as isolating a compromised endpoint, blocking malicious processes, or restoring affected files, thereby limiting the spread of the attack within the network. This proactive and intelligent approach makes EDR an indispensable component for strengthening the security posture of a modern SOC. In [197], the study presents the Sentinel-one and Sand-blast Anti Ransomware tools, which include AI for threat detection, along with a presentation of recent works related to AI-based ransomware detection. However, a major limitation in the adoption and advancement of such technologies in academic and experimental settings is that most EDR and Extended Detection and Response (XDR) solutions are proprietary and not open source. This restricts researchers' ability to access, analyze, or customize these tools for developing and testing innovative detection models, thus slowing progress in cybersecurity research and development. This critical observation opens promising avenues for future research, particularly in the development of open-source alternatives that support transparency, customization, and collaborative innovation in threat detection systems. Moreover, SIEM platforms can collect logs from both IDS and EDR systems, depending on the selected SIEM solution and the configuration of the agents responsible for log collection across the monitored infrastructure. Another key element in the SOC concept is the automation of incident response, which is enabled by SOAR (Security Orchestration, Automation, and Response) platforms. These tools empower security teams to orchestrate multiple security technologies, automate repetitive and

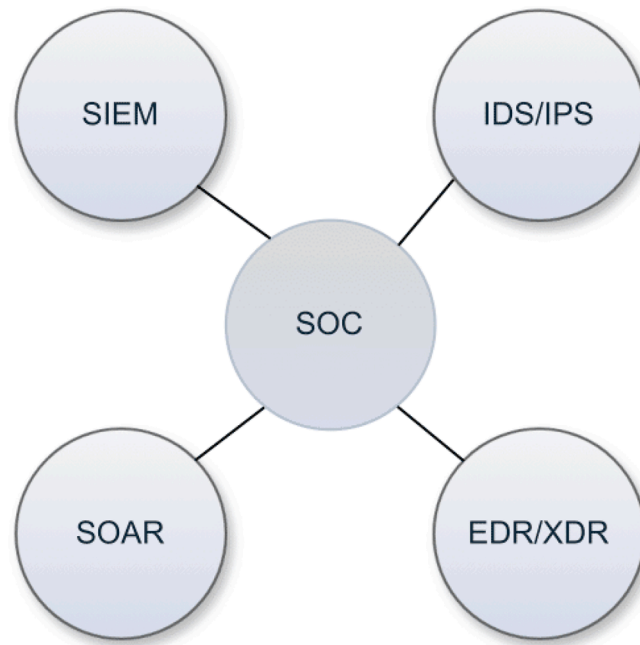


Fig. 19. Essential tools for modern healthcare threat management.

time-consuming tasks, and streamline incident response processes. By reducing the need for manual intervention, SOAR platforms help minimize human workload and enhance the overall speed and efficiency of security operations [198]. Fig. 19 displays the core cybersecurity components (SIEM, SOC, IDS/IPS, SOAR, EDR/XDR) in an integrated operational workflow.

8. Discussion, challenges, and future research directions

Although artificial intelligence has demonstrated strong potential for intrusion and anomaly detection in IoMT environments, several technical limitations and open research challenges must still be addressed to ensure reliable, secure, and large-scale deployment. The main future research directions are summarized as follows:

- **Lightweight and Resource-Efficient Intrusion Detection Frameworks.** Conventional IDS solutions often require significant computational and memory resources, making them unsuitable for constrained IoMT devices. Therefore, the design of lightweight and energy-efficient IDS models remains a primary research priority. Despite existing efforts, several challenges persist:
 - Developing lightweight ML/DL algorithms that ensure high detection accuracy while requiring minimal computational, memory, and energy resources for real-time deployment in critical IoMT applications.
 - Ensuring robustness against adversarial attacks targeting ML/DL models, which may manipulate input data or model behavior and lead to incorrect decisions. Strong security mechanisms are essential to preserve the confidentiality, integrity, and availability of sensitive medical data.
 - Supporting the heterogeneity of IoMT environments by considering diverse communication protocols, hardware platforms, operating systems, and network infrastructures to guarantee accurate and consistent attack detection.
 - Enabling distributed learning through FL, where local model training is performed on IoMT devices and only model parameters are transmitted for aggregation. When device resources are extremely limited or model confidentiality must be preserved, split learning can represent a complementary alternative.
- **Advanced and Alternative AI Techniques for IDS.** Future IDS solutions should leverage more flexible and adaptive learning paradigms to cope with the evolving nature of cyber threats:
 - Semi-supervised learning can significantly reduce the dependence on labeled data by exploiting the structural properties of IoMT traffic to detect anomalies based on limited labeled samples.
 - Self-supervised learning represents a promising solution for large-scale IoMT environments where manual data annotation is expensive and impractical. It enables the extraction of meaningful representations from massive unlabeled datasets.
 - Temporal Convolutional Networks (TCNs) constitute an efficient alternative to LSTM and RNN architectures for time-series intrusion detection, offering faster training, better parallelization, and strong capability for modeling long-range temporal dependencies at lower computational cost.
- **Cloud, Edge, and Secure Deployment Architectures.** Cloud-based and hybrid Cloud-Fog-Edge architectures provide a scalable and resilient deployment framework for IoMT security. These infrastructures offer secure online execution, real-time monitoring,

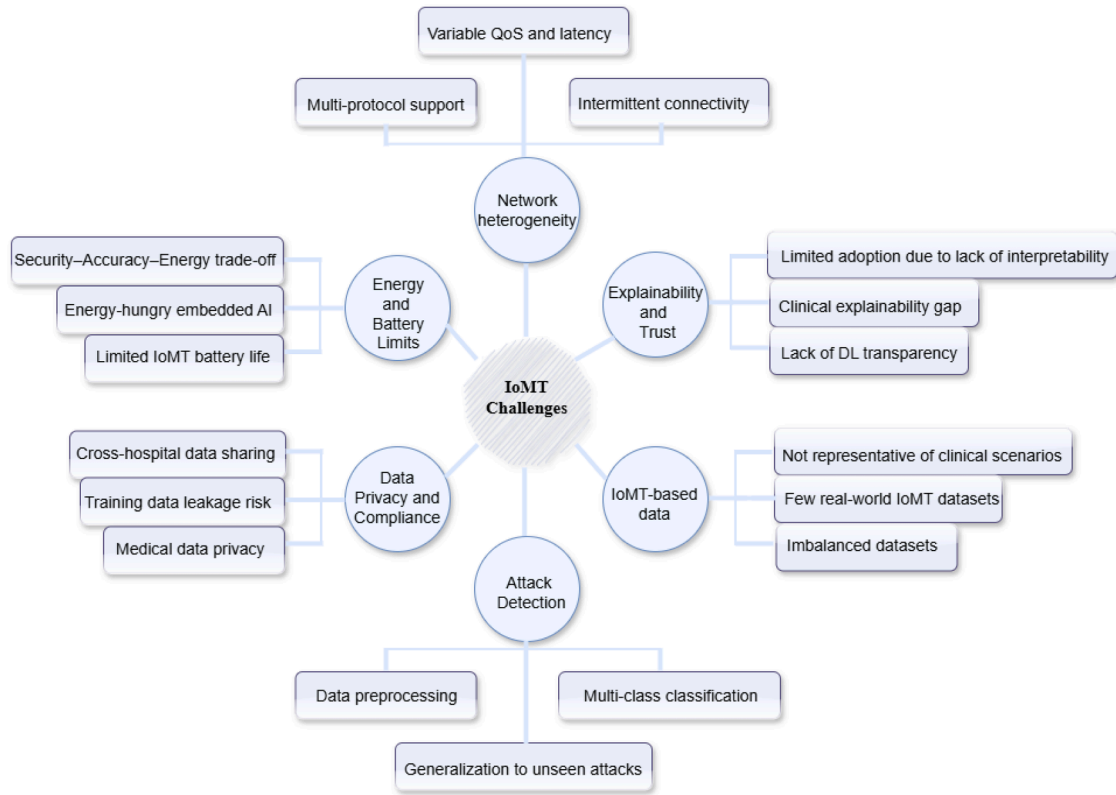


Fig. 20. IoMT-based IDS challenges.

and protection against model poisoning and data manipulation attacks. Commercial cloud platforms such as AWS, Microsoft Azure, and Google Cloud integrate advanced security services (encryption, access control, audit logging, and regulatory compliance), which enhance system availability, fault tolerance, and data protection.

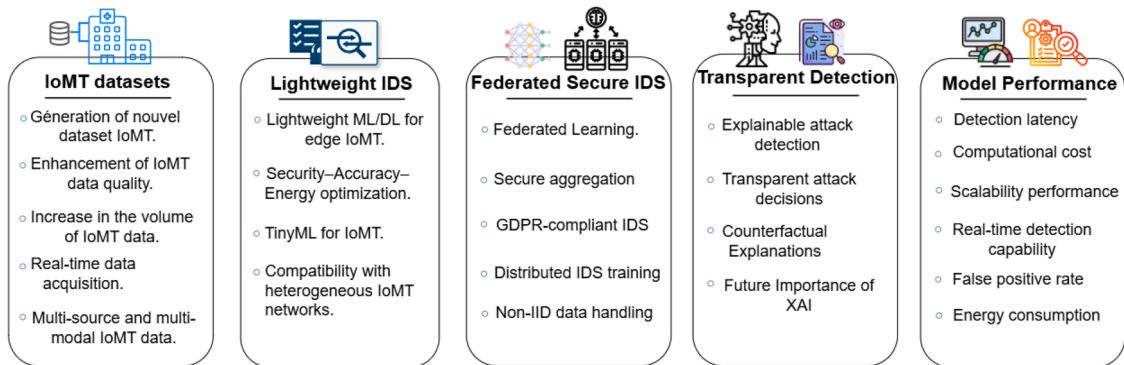


Fig. 21. Future research directions for IoMT-based IDS.

- **Large-Scale and Realistic IoMT Security Datasets.** The absence of comprehensive and realistic datasets remains a major obstacle for validating IDS solutions in medical environments:
 - Future datasets should integrate multiple communication technologies and protocols beyond MQTT and CoAP, such as Zigbee, LoRa, NB-IoT, 5G/6G, and Kafka-based streaming systems.
 - Existing datasets often focus only on patient-server communication, while ignoring critical security components such as firewalls, proxies, IDS/IPS, and SIEM platforms. Including these elements would significantly improve attack traceability and system realism.
 - Only a limited number of datasets, such as CICIoMT2024, provide multi-class attack labeling. There is a strong need for datasets covering dozens of attack categories to support robust multi-class and zero-day attack detection.

- Lightweight IDS for embedded systems must consider attacks related to communication buses, memory corruption, physical tampering, and abnormal energy consumption. However, dedicated datasets addressing these attack categories remain scarce.

Figs. 20 and 21 present the main technical challenges and promising future research avenues for strengthening the security of IoMT-based intrusion detection systems.

9. Conclusion

This study has presented a comprehensive analysis of the IoMT architecture, detailing its layered structure and emphasizing the specific security vulnerabilities associated with each component. Particular attention has been paid to the types of cyberattacks that can compromise the integrity, confidentiality, and availability of medical data across the different layers, from devices to cloud-based infrastructures. The survey has explored the full pipeline involved in the development of AI-based IDS tailored for IoMT environments. This includes the phases of data preprocessing, feature engineering, model training, evaluation, and deployment. A thorough literature review of ML and DL techniques applied to intrusion detection in healthcare-related systems has been conducted, showcasing their strengths and limitations. Additionally, the paper has identified and categorized the most widely used public datasets in IoMT and IoT security research, providing valuable insights into their characteristics, relevance, and applicability. These datasets play a crucial role in benchmarking the performance of IDS models and understanding the real-world implications of cyberattacks targeting healthcare institutions. By outlining the challenges and constraints faced when securing IoMT systems including data scarcity, class imbalance, device heterogeneity, and resource limitations this study underscores the need for adaptive, lightweight, and context-aware security mechanisms. Future research should continue to explore hybrid detection approaches, FL strategies, and edge-level intelligence to enhance the responsiveness and resilience of IoMT security frameworks.

CRedit authorship contribution statement

Mohammed Yacoubi: Writing – review & editing, Writing – original draft; **Omar Moussaoui:** Validation, Supervision; **Cyril Drocourt:** Validation, Supervision.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

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